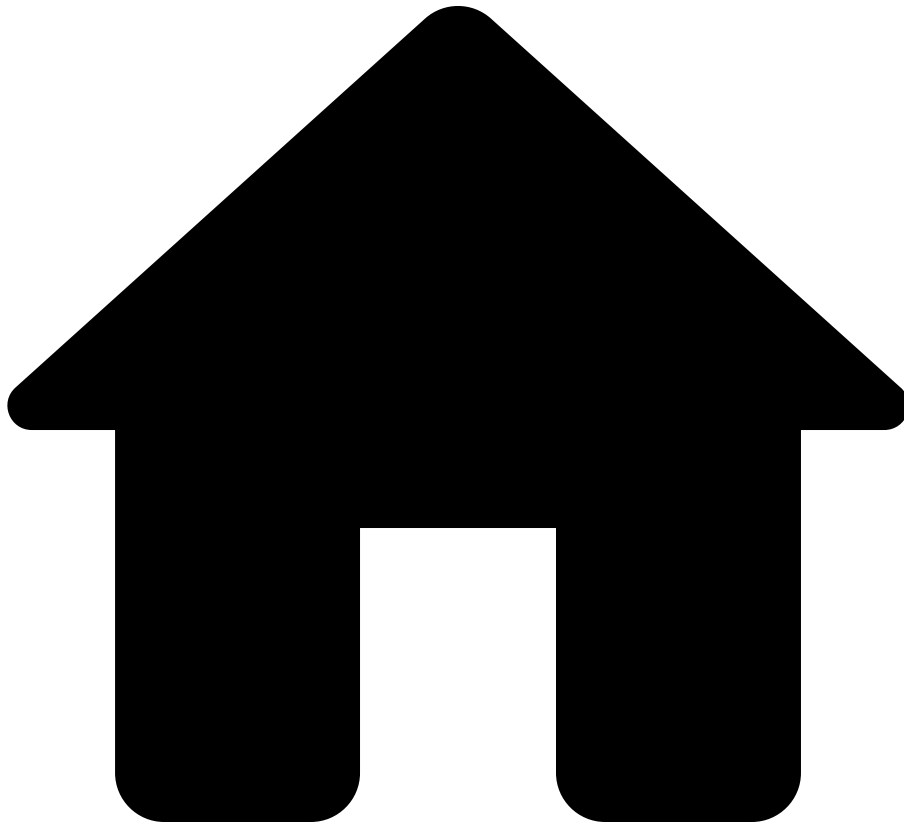


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- Settings
- [Accounts](#)
- Access

On this page

Access

The **Access** page is a comprehensive hub for managing role-based access controls. This page centralizes all the settings and tools needed to efficiently control user permissions, ensuring secure and organized access to network resources.

The Access page can be accessed by navigating to **Settings > Accounts > Access**.

The page is divided into three main tabs:

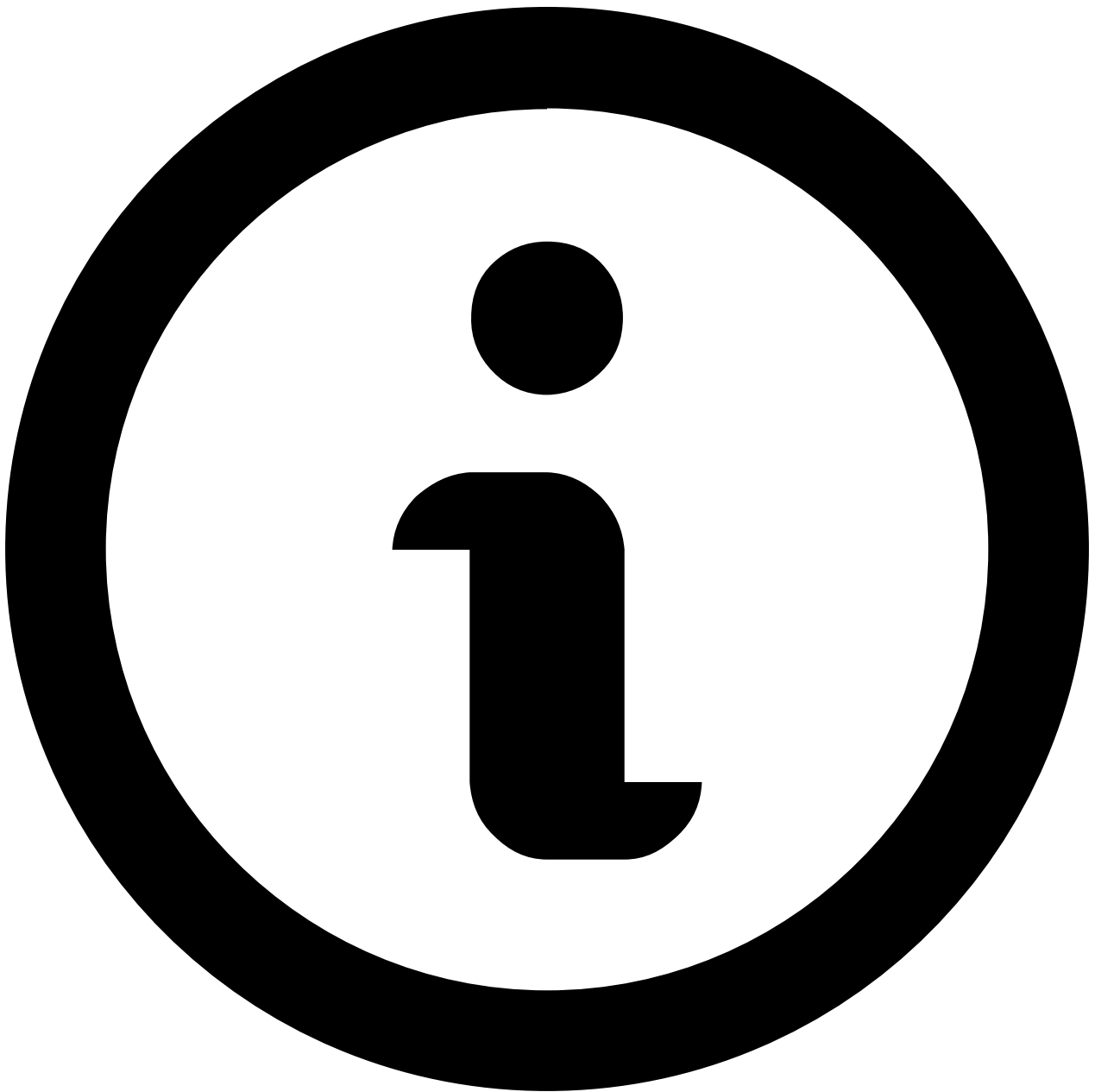
- **User Accounts:** Manage individual user accounts and their respective roles and permissions.
- **Access Control Groups:** Define and manage groups for automatic role assignment based on LDAP or SSO authentication.
- **Device Access Labels:** Control access to specific devices through customizable labels based on organizational access preferences.

The screenshot displays the 'Access' management interface. On the left, a sidebar contains navigation options: Personal, Accounts (with sub-items: Access, Login, Notices, External authentication), and System. The main content area is titled 'Access' and features three tabs: 'User accounts (3)', 'Access control groups (3)', and 'Device access labels (1)'. The 'User accounts' tab is active, showing a table of users. The table has columns for 'Email address', 'Org Admin', 'Assigned networks', 'Assigned roles', and 'Last active'. Three users are listed: 'admin@forwardnetworks.com', 'fwd-org-admin@forwardnetworks.com', and 'noreply@forwardnetworks.com'. The 'fwd-org-admin' user is highlighted. Above the table, there are filters for each column and a '3 users' indicator. A 'Set organization access preference' button is located at the top right of the page.

Email address	Org Admin	Assigned networks	Assigned roles	Last active
admin@forwardnetworks.com	Admin	Forward Forward Networks Intranet Node	Net Admin Net Admin Net Admin	1h ago
fwd-org-admin@forwardnetworks.com	Admin	Forward Forward Networks Intranet Node	Net Admin Net Admin Net Admin	1min ago
noreply@forwardnetworks.com	Admin	Forward Forward Networks Intranet Node	Net Admin Net Admin Net Admin	13h ago

Setting Organization Access Preference

Click **Set Organization Access Preference** at the top of the page to configure the default permissions for roles within the organization.



note

- The option is only available to Org Admins.
- Org Admins always have full access to all devices.
- Limited read-only roles never have access to device configurations.

Select from one of the two options provided:

- **Allow Config Access:** Configures access to all users by default. Device access must be configured to deny access based on the assigned user group.

- **Block Config Access:** Configures access to be blocked to all users by default. Device access must be configured to allow access based on the assigned user group.

The screenshot shows the 'Access' configuration page in the Forward Networks management console. The page is divided into three main sections: 'User accounts (4)', 'Access control groups (2)', and 'Device access labels (0)'. The 'User accounts' section is active, displaying a table of user accounts and their assigned networks.

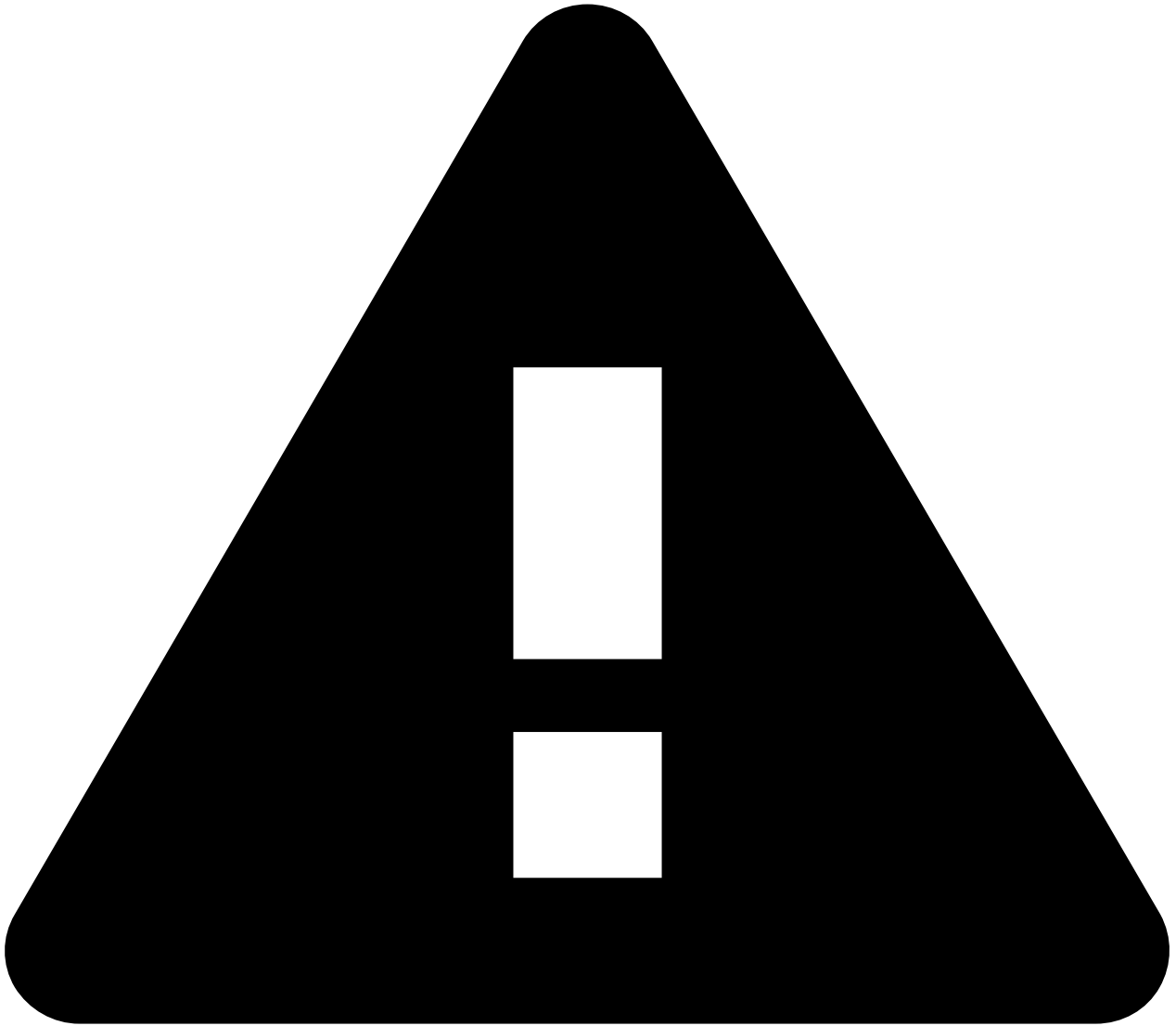
Email address	Org Admin	Assigned networks
Filter	Filter	Filter
admin@forwardnetworks.com	Admin	Forward Forward Networks Intranet Node
fwd-net-ops@forwardnetworks.com		Forward Networks
fwd-org-admin@forwardnetworks.com	Admin	Forward Forward Networks Intranet Node
noreply@forwardnetworks.com	Admin	Forward Forward Networks Intranet Node

The 'Set organization access preference' modal window is open on the right. It shows the 'Default permissions for roles' section with the following options:

- Net admin, Net operator, Read-only
- Allow config access** (selected)
- Block config access

A note below the options states: 'Organization Admins: Always have full access to all devices. Limited Read-Only Roles: Never have access to device config.'

At the bottom of the modal, there are 'Save' and 'Cancel' buttons.



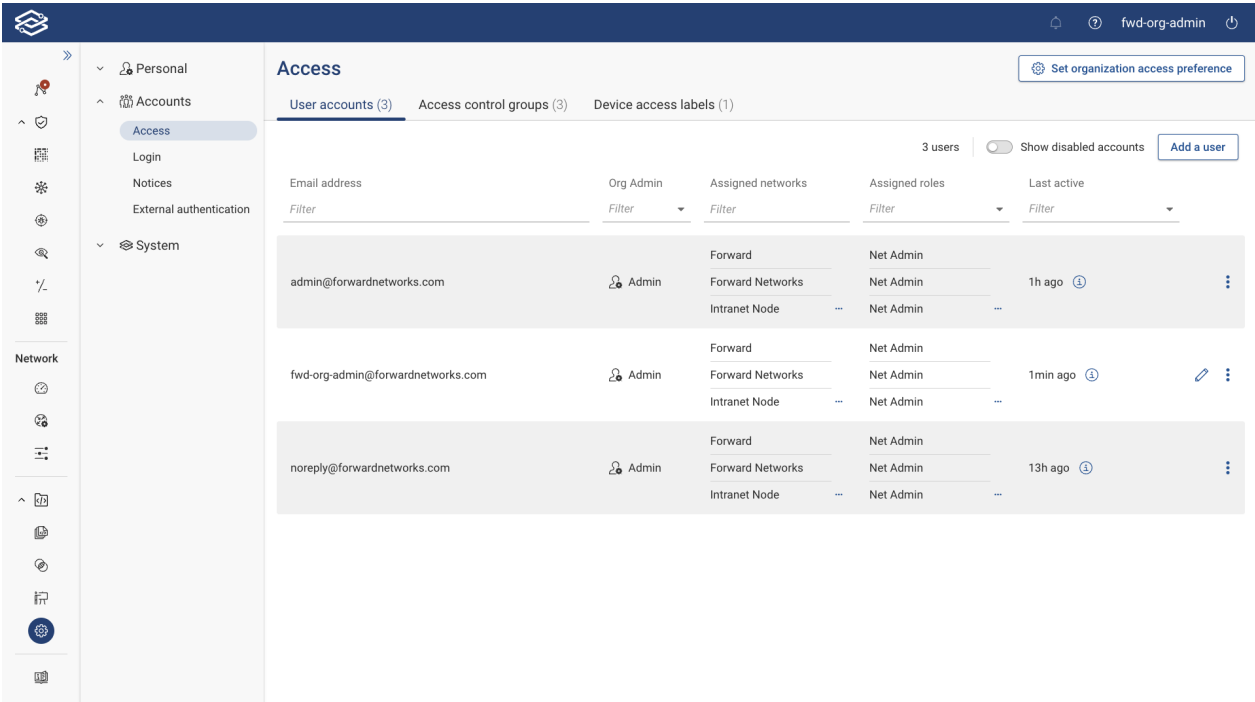
warning

Changing the default permissions for roles within the organization access preference settings will reset device configuration access and is irreversible. This action will **DELETE ALL** previously configured device access labels and **REVERT ALL** user device configuration access to the new default settings.

User Accounts

The User Accounts tab displays all currently enabled accounts, with the option to show all disabled accounts if needed. This tab displays the following information for each user account:

- **Source:** Indicates the authentication method for the user account (LDAP or SAML).
- **Email Address:** Displays the email address assigned to the user account.
- **Username:** Displays the username of a user account authenticated via LDAP or SAML.
- **Access Control Groups:** Lists the access control groups assigned to a user account authenticated via LDAP or SAML.
- **Org Admin:** Displays if the user account is assigned as an Org Admin.
- **Assigned Networks:** Displays the networks to which the user account has access.
- **Assigned Roles:** Displays the roles of the account for each assigned network.
- **Device Access Labels:** Lists the device access labels assigned to a user account authenticated via LDAP or SAML.
- **Last Active:** Displays the time since the account was last active.



Adding a User Account

Step 1: Adding a User

To add a user account, navigate to **Settings > Access > User Accounts**, and select **Add a User**.

Step 2: Configuring a new user account

From the **Add a New User** panel, enter the following information to create the user account:

- **Email:** Provide the email address to be associated with the user account.
- **Password:** Set a temporary password for the user account.
- **Enabled:** Set the user account to be in a disabled or enabled state once provisioned.
- **Used by Forward Support:** Enable this option if this account will be used by Forward Networks Support.
- **Org Admin:** Set the user account as an organization admin if enabled.
- **Network Roles:** Select one or more networks to provide access to and select the role the user account will be assigned for each selected network. User accounts set as Org Admins will have access to all networks by default.

Click **Save** to finish provisioning the user account.

The screenshot shows the 'Add a new user' panel in the Forward Networks admin interface. The panel is titled 'Add a new user' and has a close button (X) in the top right corner. It contains the following fields and options:

- Email:** A text input field with the value 'fwd-net-ops@forwardnetworks.com'.
- Password:** A password input field with a masked password '.....' and an eye icon to toggle visibility.
- Enabled:** A toggle switch that is currently turned on.
- Used by Forward Support:** A toggle switch that is currently turned off, with a help icon (?) next to it.
- Organization Admin:** A toggle switch that is currently turned off.
- Network roles:** A section with two dropdown menus:
 - Network:** A dropdown menu with 'Forward Networks' selected.
 - Role:** A dropdown menu with 'Net Operator' selected, and a help icon (?) next to it.
- + Add another network role:** A link to add more network roles.
- Save:** A blue button to save the user account.
- Cancel:** A white button with a blue border to cancel the operation.

The background of the screenshot shows the 'Access' section of the admin interface, with a table of user accounts. The table has columns for 'Email address', 'Org Admin', and 'Assigned networks'. The table contains three rows of data:

Email address	Org Admin	Assigned networks
admin@forwardnetworks.com	Admin	Forward Forward Networks Intranet Node
fwd-org-admin@forwardnetworks.com	Admin	Forward Forward Networks Intranet Node
noreply@forwardnetworks.com	Admin	Forward Forward Networks Intranet Node

Managing User Accounts

User accounts are displayed and can be managed from the User Accounts tab; the following options are available at the end of a user account displayed:

- **Edit:** Selecting the edit icon opens the **Edit User** pane for editing a user account's settings.
- **Additional Options:**
 - **Disable:** To disable a user account, click the additional options icon and select **Disable**.
 - **Delete:** To delete a user account, click the additional options icon and select **Delete**.

The screenshot shows the 'Access' page in the Forward Networks interface. The left sidebar contains navigation options: Snapshot, Personal (Preferences, Account, Notifications), Accounts (Access, Login, Notices, External authentication), Network, and System (Org preferences, Integrations, Webhooks, Notification settings, Privacy Policy). The main content area is titled 'Access' and includes tabs for 'User accounts (4)', 'Access control groups (2)', and 'Device access labels (1)'. Below the tabs, there's a header with '4 users', a toggle for 'Show disabled accounts', and an 'Add a user' button. The table below lists user accounts with columns for Email address, Org Admin, Assigned networks, Assigned roles, and Last active. The table contains four rows of user data.

Email address	Org Admin	Assigned networks	Assigned roles	Last active
admin@forwardnetworks.com	Admin	Forward Forward Networks Intranet Node	Net Admin Net Admin Net Admin	5h ago
fwd-netops@forwardnetworks.com		Forward Networks	Net Operator	n/a
fwd-org-admin@forwardnetworks.com	Admin	Forward Forward Networks Intranet Node	Net Admin Net Admin Net Admin	4min ago
noreply@forwardnetworks.com	Admin	Forward Forward Networks Intranet Node	Net Admin Net Admin Net Admin	5h ago

Disabling Inactive Accounts

This feature allows Org Admins to set a limit on the number of days an account can remain inactive before being automatically disabled. This setting applies to all roles except Org Admins. Disabled accounts can be manually re-enabled by an Org Admin or by using Forward's APIs.

To enable this feature, navigate to **Settings > System > Org Preferences**, and within the **Account Management** field, enter the maximum number of inactive days for an account before it is disabled.

Org preferences

These settings affect all user accounts in your organization

Account management Maximum period of account inactivity days [?](#)

Passwords	Minimum password lifetime	0 hours
	Maximum password lifetime	2147483647 days

Time zone Organization's preferred time zone
(GMT+00:00) Coordinated Universal Tim... ?

Displayed time zone
The time zone used to display times in this UI

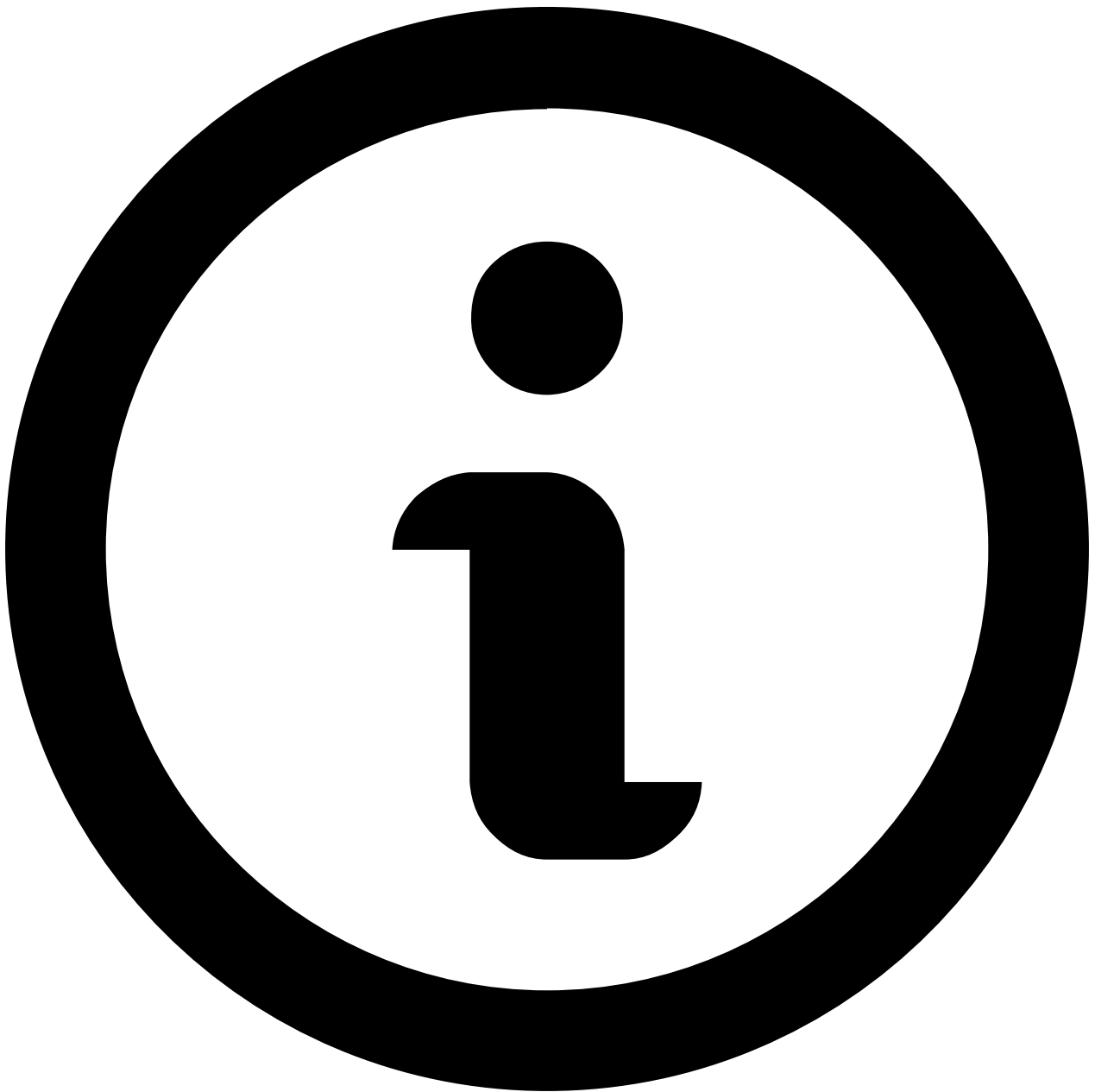
☒ Organization's preferred time zone
(GMT+00:00) Coordinated Universal Time - UTC

☐ Viewer's local time zone

Credential copying [?](#) ☐ Copy credentials between networks [?](#)

☐ Enable credential copying for Network Operators [?](#)

Workflows networks Enable auto propagation of user roles



note

Any account that interacts with the system, whether through the **UI** or **API requests**, is considered active. This includes service accounts that exclusively perform API operations.

Manually Enabling Disabled Accounts [â](#)

To manually enable an account, navigate to **Settings > Access > User Accounts**, toggle the **Show Disabled Accounts** option to display all disabled accounts, and enable a

disabled account by selecting the kebab icon to display the additional option to enable the account.

The screenshot shows the Forward Networks Admin console. The left sidebar has a 'Personal' section with 'Access' highlighted. The main area is titled 'Access' and shows a table of user accounts. The 'Show disabled accounts' toggle is turned on. A dropdown menu is open for the 'fwd-org-admin@forwardnetworks.com' account, showing 'Enable' and 'Delete' options.

Email address	Org Admin	Assigned networks	Assigned roles	Last active
admin@forwardnetworks.com	Admin	Forward Forward Networks Intranet Node	Net Admin Net Admin Net Admin	5h ago
fwd-netops@forwardnetworks.com		Forward Networks	Net Operator	n/a
fwd-org-admin@forwardnetworks.com	Admin	Forward Forward Networks Intranet Node	Net Admin Net Admin Net Admin	5min ago
noreply@forwardnetworks.com	Admin	Forward Forward Networks Intranet Node	Net Admin Net Admin Net Admin	5h ago

Alternatively, selecting the edit icon next to the account and toggling the enabled setting from the Edit User window will also enable the account.

The screenshot shows the 'Edit user' window for the 'fwd-org-admin@forwardnetworks.com' account. The 'Enabled' toggle is turned on. The 'Network roles' section shows the 'Forward Networks' network and the 'Net Operator' role.

Email address	Org Admin	Assigned networks
admin@forwardnetworks.com	Admin	Forward Forward Networks Intranet Node
fwd-netops@forwardnetworks.com		Forward Networks
fwd-org-admin@forwardnetworks.com	Admin	Forward Forward Networks Intranet Node
noreply@forwardnetworks.com	Admin	Forward Forward Networks Intranet Node

API Token Management

Org Admins can view and manage all user-created API tokens from the **User Accounts** tab in **Accounts > Access**. This view includes the columns **API Token First Created** and **API Token Last Used**, providing insight into token activity across the organization.

To view or delete individual tokens, click the **Edit** icon next to a user to open the **Edit User** drawer. The **API Tokens** section lists all tokens created by that user and includes:

- **Access Key**
- **Creation Date**
- **Expiration Date** (only shown if an expiration policy has been set)
- **Last Used**

Accounts

Access

0 selected

Enable

Disable

Email address

Filter

admin@forwardnetworks.com

fwd-netops@forwardnetworks.com

noreply@forwardnetworks.com

operator@forwardnetworks.com

org-admin@forwardnetworks.com

Edit user

Enter a new password

Enabled

Used by Forward Support

Organization Admin

Network

Forward Networks

Role

Net Operator

Add another network role

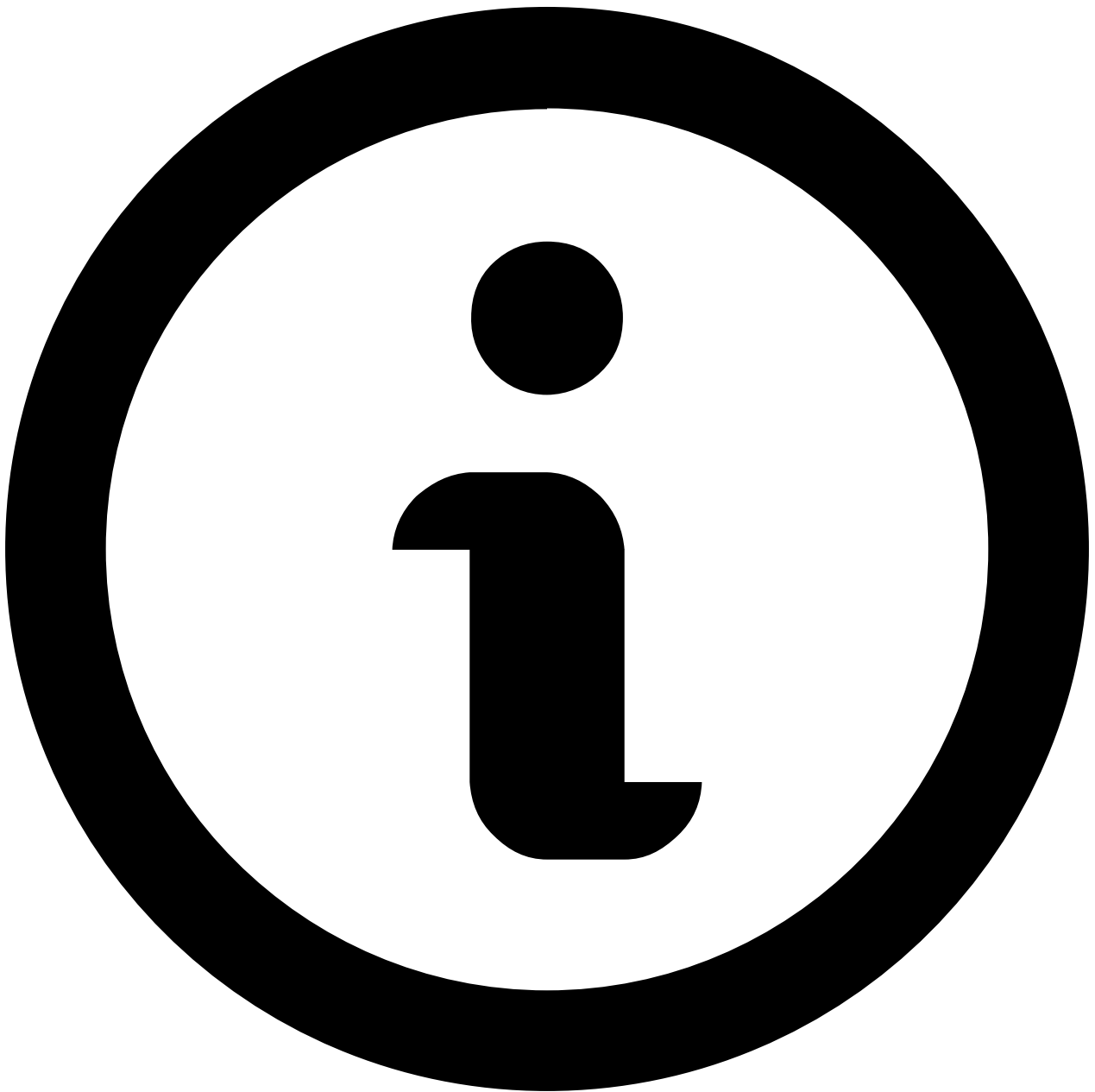
API tokens

2 keys

Access key	Creation date	Expiration date	Last used
Filter	Filter	Filter	Filter
aub4-ft3x4-bpu4	2025-05-21 08:52:59 UTC	2025-08-19 08:52:59 UTC	never
qh37-jeqpa-4222	2025-05-21 08:53:18 UTC	2025-08-19 08:53:18 UTC	never

Save

Cancel

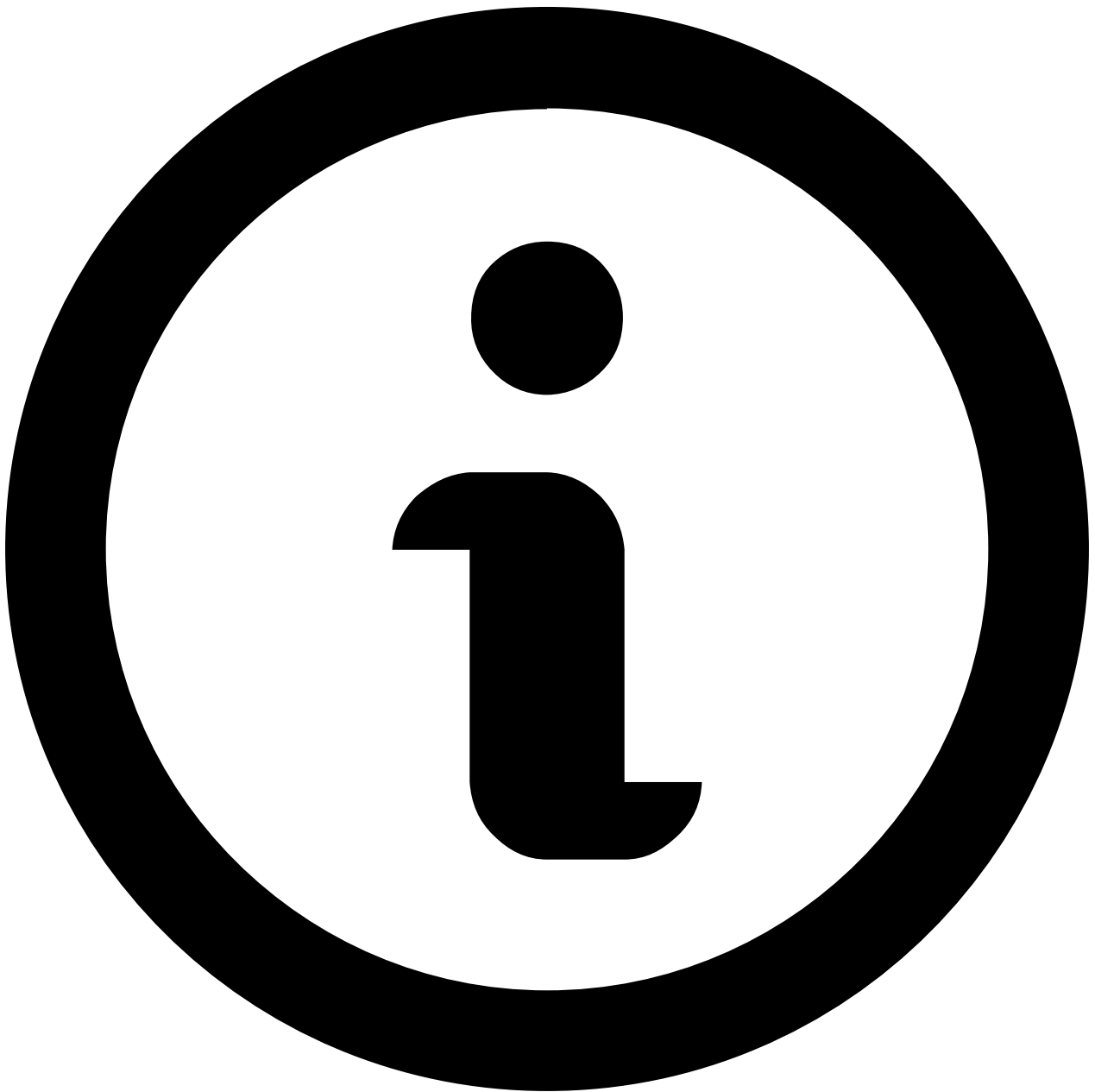


note

Org Admins can delete any token listed but cannot create tokens on behalf of other users.

Token expiration is controlled by the **API token maximum lifetime** setting in [Org Preferences](#). If configured, tokens expire automatically after the specified number of days. If the field is left blank, tokens do not expire. Expired tokens cannot be reactivated, and any updates to the lifetime setting only affect tokens that have not yet expired.

[illegible]

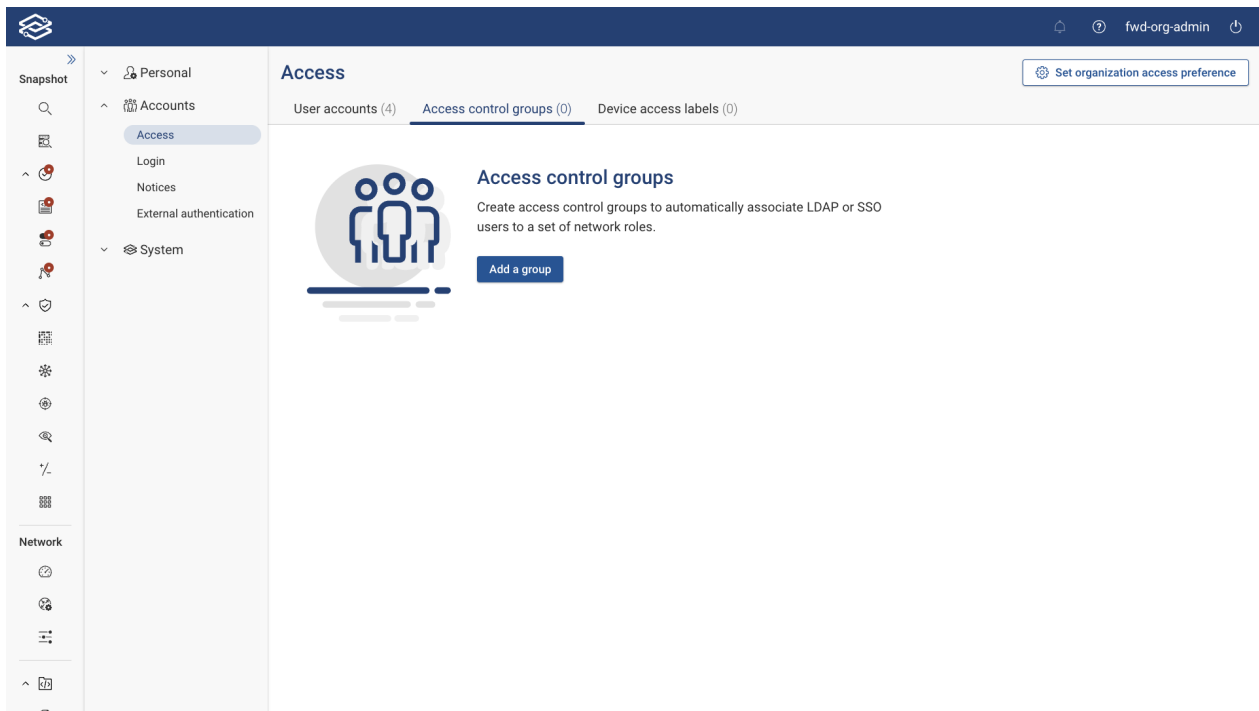


note

Expiration dates are only displayed if the **API token maximum lifetime** has been configured with a non-zero value.

Access Control Groups

Access control groups allow Org Admins to automatically associate externally authenticated LDAP or SSO users with org or network roles in the Forward Enterprise application.



Adding an Access Control Group

To add an access control group, navigate to **Settings > Access > Access Control Groups**, and select **Add a Group**. Provide the following information:

- **Access Control Group Name:** A unique name for the access control group.
- **LDAP/SSO Groups:** One or more LDAP/SSO groups. Enter a group name, then click **Add Another LDAP/SSO Group** if you need to add more. Note that group matching is case-insensitive.
- **Org Admin:** Toggle to set the access control group as an Org Admin.

Network Roles:

- **Network:** One or more networks to assign the user to with the role selected above. Select a network from the menu, then click **Add Another Network** if you need to add more.

- **Role:** The Forward Enterprise role to associate with the LDAP/SSO users that belong to any of the LDAP/SSO groups listed above. The available roles include Net Admin, Net Operator, Read-Only and Limited Read-Only.

Device Config Access:

- **Block Config Access on Labeled Devices (optional):** This option will be available if the Allow Config Access setting was set as the default permission for roles from Set Organization Access Preference. Select the device access label(s) from the list to assign it to the access control group.
- **Allow Config Access on Labeled Devices (optional):** This option will be available if the Block Config Access setting is set as the default permission for roles from Set Organization Access Preference. Select the device access label(s) from the list to assign it to the access control group.

The screenshot shows the Forward Enterprise web interface. The left sidebar contains navigation options: Snapshot, Personal (Accounts, Access, Login, Notices, External authentication), and System. The main content area is titled 'Access' and includes tabs for 'User accounts (4)', 'Access control groups (0)', and 'Device access labels (0)'. The 'Access control groups' tab is active, displaying a graphic of three people and the text 'Create access control groups to automatically associate LDAP users to a set of network roles.' with an 'Add a group' button. A modal titled 'Add an access control group' is open on the right. It contains the following fields and options:

- Access control group name:** A text input field with 'read-only' entered.
- LDAP/SSO groups:** A section with a 'Group name' input field containing 'readonly' and a '+ Add another LDAP/SSO group' link.
- Organization Admin:** A toggle switch currently turned off.
- Network Roles:** A section with a 'Network' dropdown set to 'Forward Networks' and a 'Role' dropdown set to 'Read-Only'. A '+ Add another network role' link is below.
- Device config access:** A section with a blue informational box stating: 'By default, all roles are blocked from accessing device configuration. Manage the preference. You can allow this access control group from accessing device'. Below this box are 'Save' and 'Cancel' buttons.

- **Preview Device Config Access Policy:** Clicking this displays all devices and their config access in association with the selected Access Control Group for each of the assigned networks.

Snapshot

Personal

Accounts

System

Preferences

Account

Notifications

Access

Login

Notices

External authentication

Org preferences

Integrations

Webhooks

Notification settings

Privacy Policy

Access

User accounts (4)

Access con

0 selected

Edit device access

Name

Filter

LDAP

netops

read-only

netops

readon

Device config access policy: read-only

Forward Networks

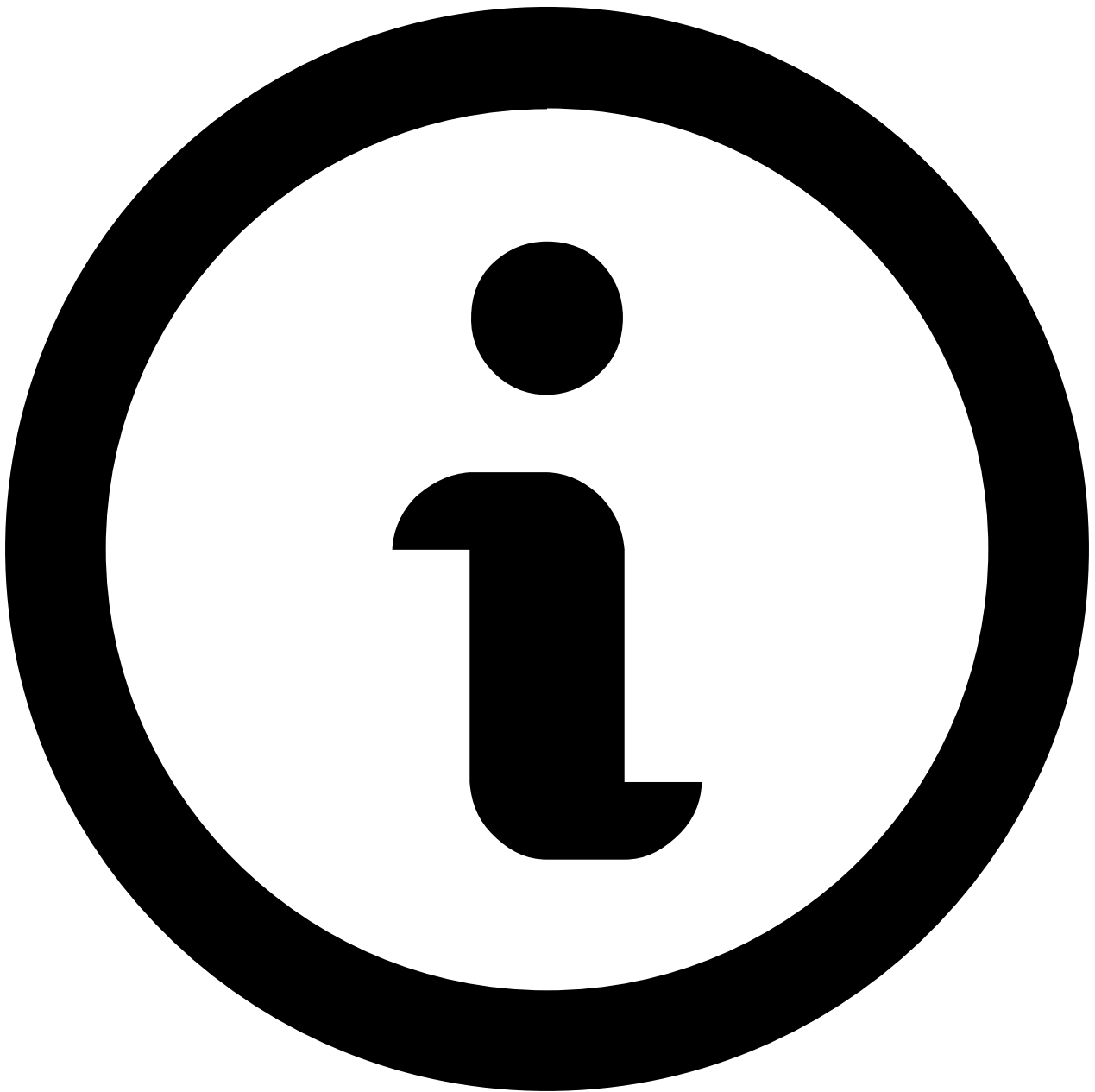
Read-Only

15 devices

Devices	IP/host	Blocked by	Config access
Filter	Filter	Filter	Filter
sw-dell-01	10.128.65.132	None	Allowed
cp01	10.128.64.42	readonly	Blocked
n5k02	10.128.64.53	readonly	Blocked
n5k01	10.128.64.48	readonly	Blocked
a7050	10.128.64.17	None	Allowed
n9k01	10.128.64.14	readonly	Blocked
cat6k	10.128.64.12	None	Allowed
n7k01	10.128.64.47	readonly	Blocked
qfx01	10.128.64.16	readonly	Blocked
bigip3k1	10.128.64.46	readonly	Blocked

Back to group details

Select **Save** when ready to finish adding the group.



note

For SSO Groups: The SAML assertion must include a group attribute statement with an attribute name that includes the `group` (e.g., `group` , `groups` , `my-groups`), as in the example from Okta below:

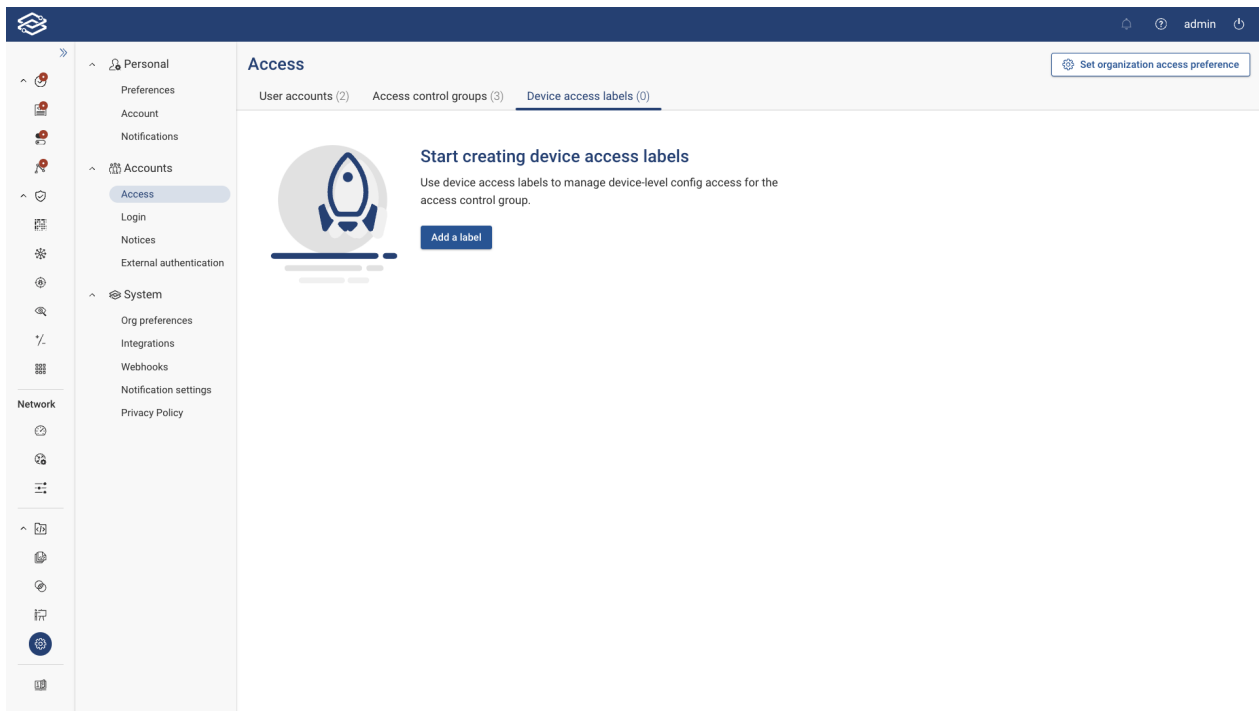
Okta Admin console interface showing the configuration for Attribute Statements (optional). The configuration includes a table with columns: Name, Name format (optional), and Value. The 'Name' field is set to 'groups', 'Name format' is 'Unspecified', and 'Filter' is 'Matches regex'. A blue arrow points to the 'groups' field. Below this is a 'Preview the SAML assertion' section with a 'Preview the SAML Assertion' button. The footer shows copyright information for 2022 Okta, Inc.

Device Access Labels

Device Access labels determine which devices are allowed or blocked based on the organization's access preference setting.

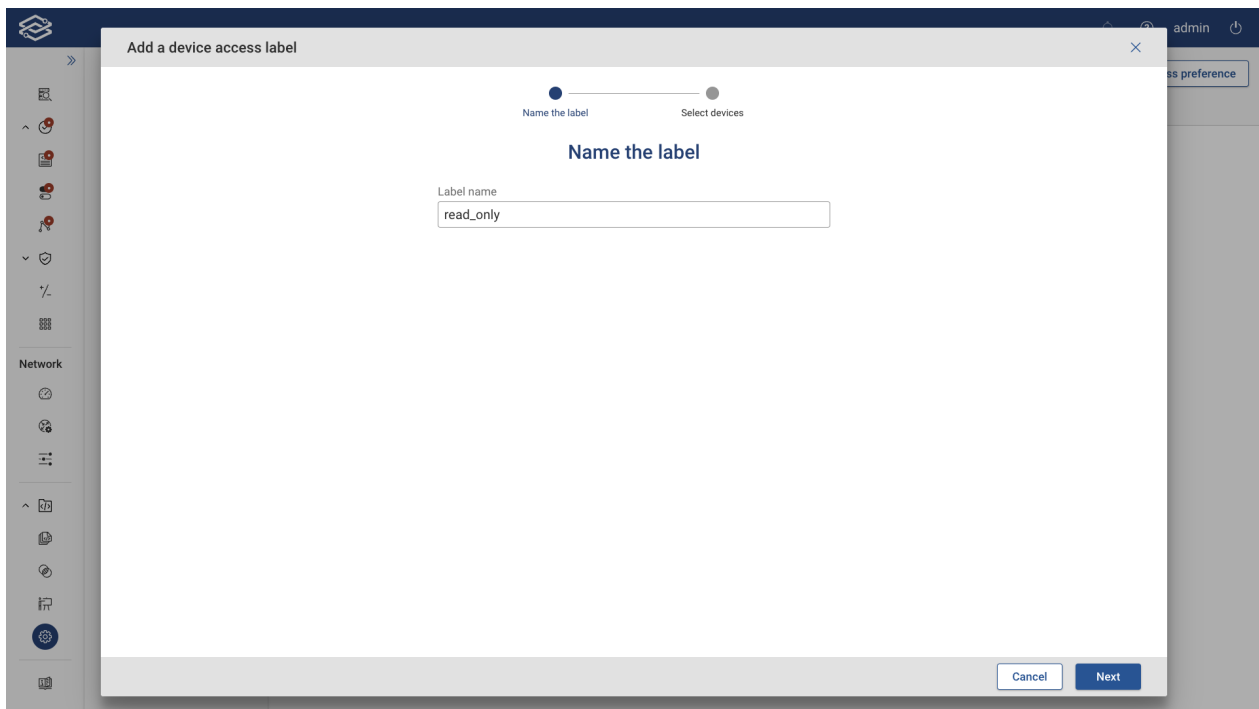
Adding a Label

To add a Device Access Label, navigate to **Settings > Access > Device Access Labels**, and select **Add a Label** to open the wizard.



Step 1: Name the Label

Provide a unique name for the label. Click **Next**.

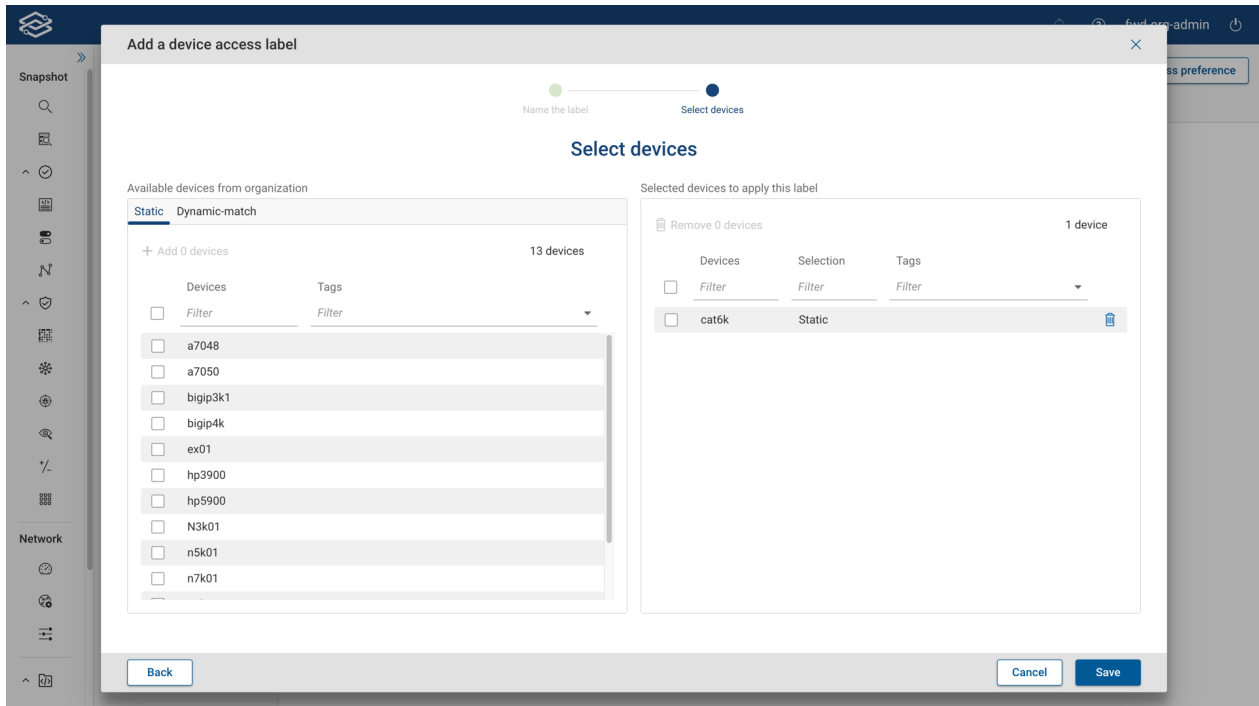


Step 2: Select Devices

Select devices to assign to the label.

- **Static:** Select individual devices by searching by device name or tag within the **static** tab.

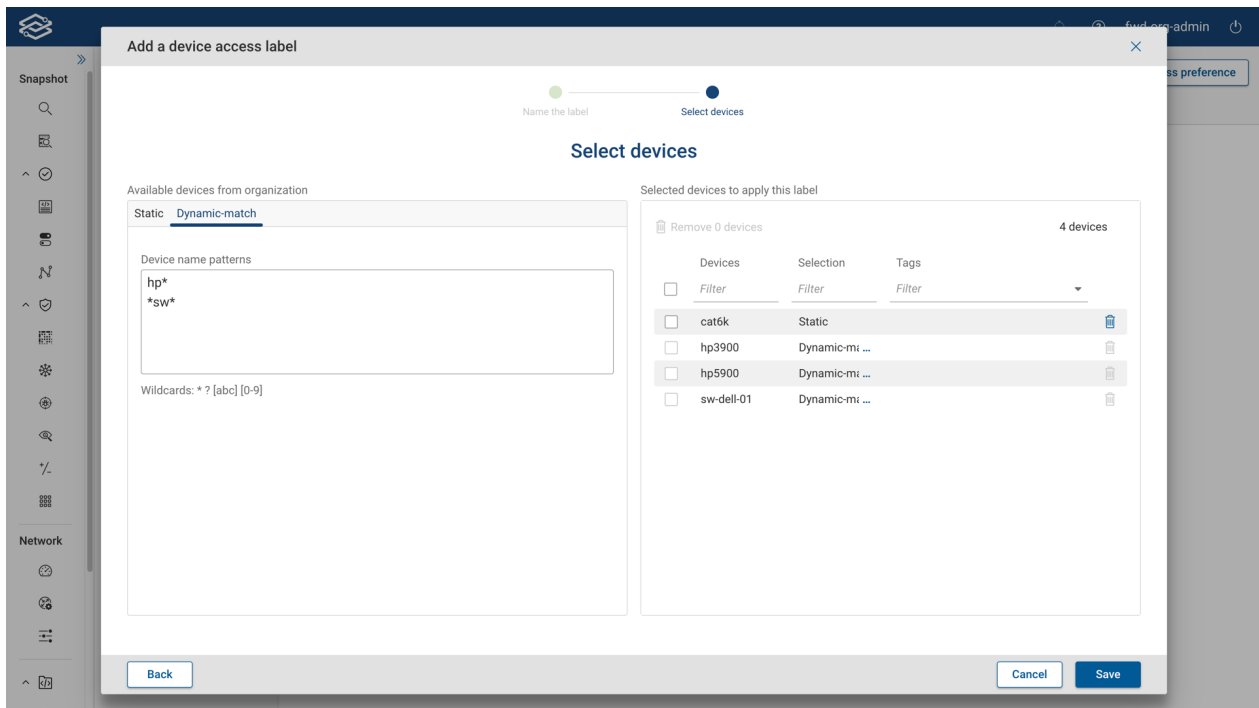
Selected devices can be removed by selecting the **delete** icon next to them.



- **Dynamic-match:** Use device name patterns with wildcards to dynamically select devices within the **dynamic-match** tab. Multiple patterns can be added, one pattern per line. Future devices matching the pattern will be automatically added to the label.

Static selection of devices takes preference over dynamic-match selection of devices.

Devices added through dynamic matching cannot be individually deleted. To delete a dynamically matched device, the associated device name pattern must be removed.



Click **Save** to finish adding the device access label.

Managing Device Access Labels

Device Access Labels are listed within the **Device Access Labels** tab. Each device access label row provides the following information:

- **Label:** Lists the name of the label.
- **Devices:** Displays the number of devices assigned to the label. Clicking the device number opens the **Edit Label Devices** tab, displaying a list of the devices and affiliated tags. Selecting **Edit Devices** opens the **Edit Devices on This Label** panel to edit the devices assigned to the label.
- **Access Control Groups:** Displays the number of access control groups to which the label is assigned. Click on the displayed access control group number to view and edit the **label-blocked** or **label-allowed** access control groups tab, depending on the org access preference settings being used. You can view the names of the access control groups assigned, assign additional groups, or remove existing groups if needed.
- **Last Edited:** Displays when the label was last edited.
- **Last Edited By:** Displays the name of the last user who edited the label.

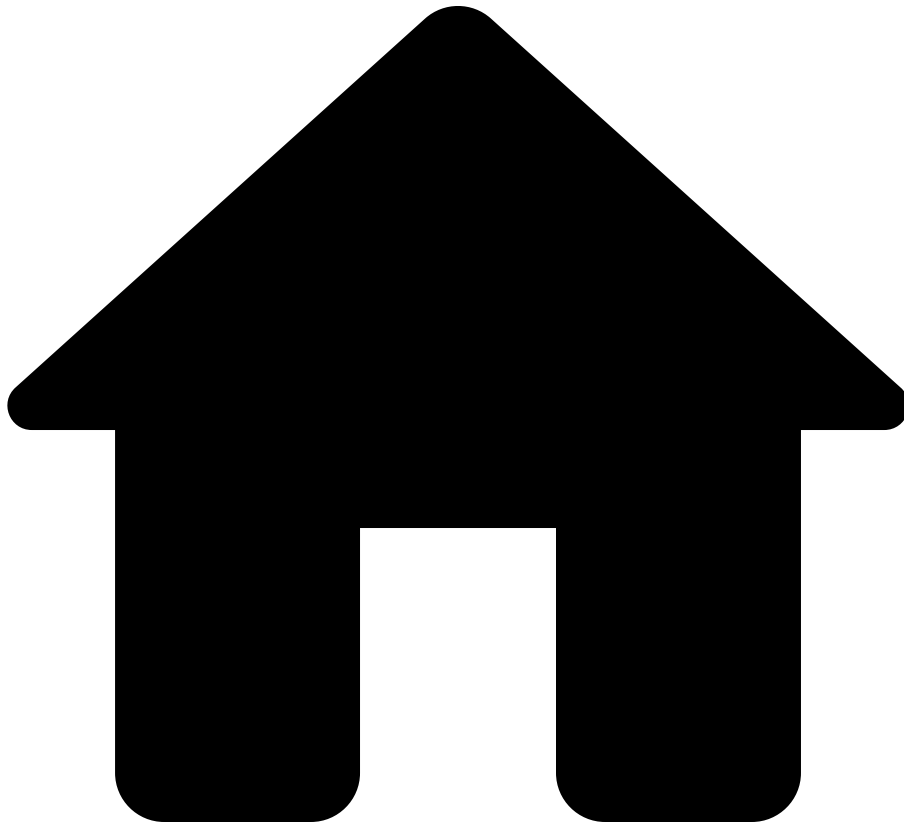
At the end of each device access label row are additional options for editing and deleting:

- **Edit:** Select the **Edit** icon to open the Edit Label panel to edit the device access label.
- **Delete:** Select the **Delete** icon to delete the selected label.

The screenshot displays the 'Access' management page in the Forward Networks dashboard. The left sidebar contains navigation links for Personal, Accounts, and System settings. The main panel is titled 'Access' and features three tabs: 'User accounts (2)', 'Access control groups (3)', and 'Device access labels (1)'. The 'Device access labels' tab is selected, showing a table with one entry, 'read_only'. The table headers are 'Label', 'Devices', 'Access control groups', 'Last edited', and 'Last edited by'. The 'read_only' label is associated with 4 devices and was last edited less than 1 minute ago by 'admin@forwardnetworks.com'. At the end of the row, there are icons for editing and deleting the label. A 'Set organization access preference' button is located in the top right corner of the main panel.

Label	Devices	Access control groups	Last edited	Last edited by
read_only	4	0	<1 min ago	admin@forwardnetworks.com

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- Settings
- [Accounts](#)
- Role-Based Access Control (RBAC)

On this page

Role-Based Access Control (RBAC)

Forward Enterprise supports Role-Based Access Control (RBAC), enabling organizations to manage user permissions with precision and align access with job responsibilities. This ensures that users only have access to the data and actions necessary for their role, improving both security and operational efficiency.

The following roles are available:

- **Org Admin:** Admin user with full access to the platform and all its settings, including Network creation, and user account management.
- **Network Admin:** A Network Admin has full access to all Forward applications for the network and can configure the network, including its collection settings.
- **Network Operator:** A Network Operator has full access to all Forward applications for the network, but they cannot configure network settings or trigger Snapshot collection.
- **Workspace Operator:** Assigned automatically when you create a workspace network (when [credential copying](#) is enabled). Has all Network Operator capabilities plus the ability to trigger snapshot collection, copy and delete sources, view credentials, proxies, and jump servers, and edit or delete snapshots and networks. Cannot add or edit devices or credentials, or upload or invalidate snapshots. See [Workspace Network RBAC](#) for details.
- **Read-only:** A Read-only user has only view (read) access to the network. They cannot create new data, nor change any settings.
- **Limited Read-only:** Limited Read-only users are restricted Read-only users, with no access to sensitive data like configuration and state files of the network devices, nor to any application that could expose them ([NQE](#), [Inventory+](#), and Security applications).

Settings

Functionality	Limited Read-Only	Read-Only	Network Operator	Workspace Operator	Network Admin	Org Admin
Personal						
Personal - Preferences	Yes	Yes	Yes	Yes	Yes	Yes
Personal - Account	Yes	Yes	Yes	Yes	Yes	Yes
Personal - Notifications	Yes	Yes	Yes	Yes	Yes	Yes
Accounts						

Functionality	Limited Read-Only	Read-Only	Network Operator	Workspace Operator	Network Admin	Org Admin
Accounts - User accounts (includes accounts creation/editing/deleting)	No	No	No	No	No	Yes
Accounts - Login (includes 2FA)	No	No	No	No	No	Yes
Accounts - Login Notice	No	No	No	No	No	Yes
Accounts - TACACS	No	No	No	No	No	Yes
Accounts - SSO	No	No	No	No	No	Yes
System						
System - General (includes session mgmt, password strength, experimental features)	No	No	No	No	No	Yes
	No	No	No	No	No	Yes

Functionality	Limited Read-Only	Read-Only	Network Operator	Workspace Operator	Network Admin	Org Admin
System - Software Update						
System - Backup / Restore	No	No	No	No	No	Yes
System - Database Import / Export	No	No	No	No	No	Yes
System - Integrations	No	No	No	No	No	Yes
System - License Key	No	No	No	No	No	Yes
System - Licensed Devices	No	No	No	No	No	Yes
System - Notification Settings	No	No	No	No	No	Yes
System - SMTP Settings	No	No	No	No	No	Yes
System - System Overview	No	No	No	No	No	Yes
System - Webhooks	No	No	No	No	No	Yes

Network

Functionality	Limited Read-Only	Read-Only	Network Operator	Workspace Operator	Network Admin	Org Admin
Devices						
Devices - View	Yes	Yes	Yes	Yes	Yes	Yes
Devices - Edit	No	No	No	No	Yes	Yes
Virtualization (NSX, ESX)						
Virtualization (NSX, ESX) - View	Yes	Yes	Yes	Yes	Yes	Yes
Virtualization (NSX, ESX) - Edit	No	No	No	No	Yes	Yes
Cloud Accounts						
Cloud Accounts - View	Yes	Yes	Yes	Yes	Yes	Yes
Cloud Accounts - Edit	No	No	No	No	Yes	Yes
Cloud Account Proxy - View	No	No	No	Yes	Yes	Yes
Cloud Account Proxy - Edit	No	No	No	No	Yes	Yes

Functionality	Limited Read-Only	Read-Only	Network Operator	Workspace Operator	Network Admin	Org Admin
Access Credentials						
Credentials - View	No	No	No	Yes	Yes	Yes
Credentials - Edit	No	No	No	No	Yes	Yes
Jump Servers - View	No	No	No	Yes	Yes	Yes
Jump Servers - Edit	No	No	No	No	Yes	Yes
Collection						
Collection Schedule - View	No	No	No	Yes	Yes	Yes
Collection Schedule - Edit	No	No	No	Yes	Yes	Yes
Collection Triggering	No	No	No	Yes	Yes	Yes
Platform						
Dashboard - View	Yes	Yes	Yes	Yes	Yes	Yes
NQE (create/edit/delete query)	No	Maybe*	Maybe**	Maybe**	Maybe**	Yes
	No	Yes	Yes	Yes	Yes	Yes

Functionality	Limited Read-Only	Read-Only	Network Operator	Workspace Operator	Network Admin	Org Admin
NQE (run query, Verify checks)						
Snapshots - Restore	No	No	Yes	Yes	Yes	Yes

* By default, No, [but permission can be granted for specific directories](#)

** By default, Yes, [but permission can be revoked for specific directories](#)

Network Snapshots (Forward Apps) [ââ](#)

Functionality	Limited Read-Only	Read-Only	Network Operator	Workspace Operator	Network Admin	Org Admin
Search						
Search	Yes	Yes	Yes	Yes	Yes	Yes
Search - Path Analysis	Yes	Yes	Yes	Yes	Yes	Yes
Search - View device configuration and state	No	Yes	Yes	Yes	Yes	Yes
Verify						
Verify - View NQE checks	No	Yes	Yes	Yes	Yes	Yes
Verify - View other checks	Yes	Yes	Yes	Yes	Yes	Yes

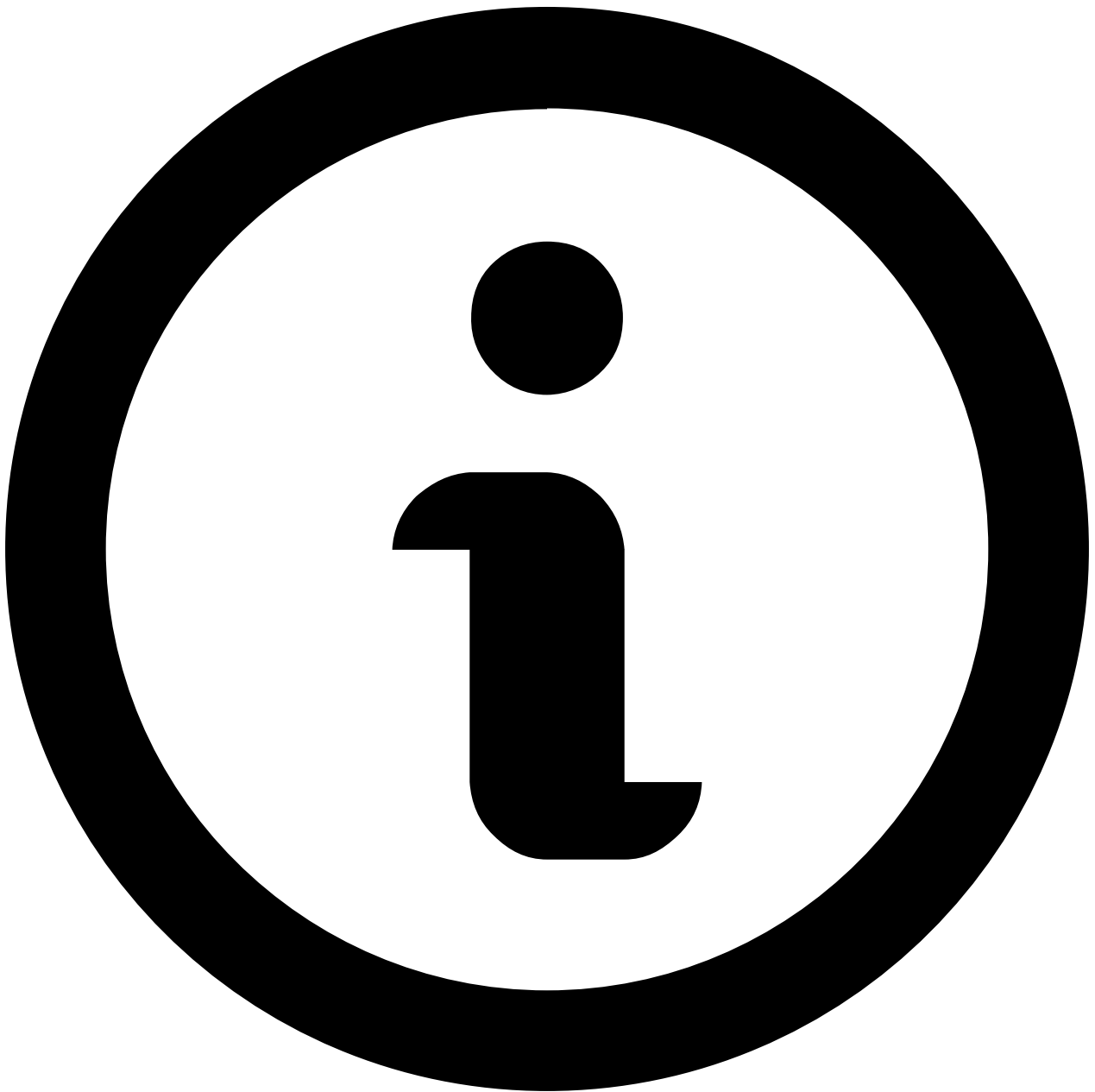
Functionality	Limited Read-Only	Read-Only	Network Operator	Workspace Operator	Network Admin	Org Admin
Verify - Add/edit/delete checks	No	No	Yes	Yes	Yes	Yes
Security						
Security Matrix - View	No	Yes	Yes	Yes	Yes	Yes
Security Matrix - Add/edit/delete	No	No	Yes	Yes	Yes	Yes
Security Blast Radius	No	Yes	Yes	Yes	Yes	Yes
Security Vulnerability	No	Yes	Yes	Yes	Yes	Yes
Security Exposure	No	Yes	Yes	Yes	Yes	Yes
Platform						
Diffs	Yes	Yes	Yes	Yes	Yes	Yes
Inventory	No	Yes	Yes	Yes	Yes	Yes
Objects - View	Yes	Yes	Yes	Yes	Yes	Yes
Objects - Add/edit/delete	No	No	Yes	Yes	Yes	Yes

Workspace Network RBAC

The Workspace network follows the role-based access control (RBAC) outlined for the parent network. The role assigned to the workspace creator depends on their role in the parent network and whether [credential copying](#) is enabled:

- **Org Admin or Network Admin in the parent network** Always receives **Network Admin** on the workspace, regardless of the credential copying setting.
- **Network Operator in the parent network, credential copying enabled** Elevated to **Workspace Operator**. This role sits between Network Operator and Network Admin. Workspace Operators can trigger and cancel collections, run connectivity tests, copy sources from the parent network, delete sources and snapshots, view credentials and jump servers (but not edit them), edit collection schedules, and edit snapshot and network metadata (name, note). They cannot add or edit devices, edit credentials, upload or invalidate snapshots, or modify network-level settings.
- **Network Operator in the parent network, credential copying disabled** Elevated to **Network Admin**, since credentials must be provided manually and cannot be copied from the parent network.

Read-only and Limited Read-only users cannot create workspace networks.



note

The Workspace Operator role applies only to newly created workspace networks. Existing workspaces are not migrated – users who were previously elevated to Network Admin on an existing workspace retain that role.

Workspace Operator Permissions[↗](#)

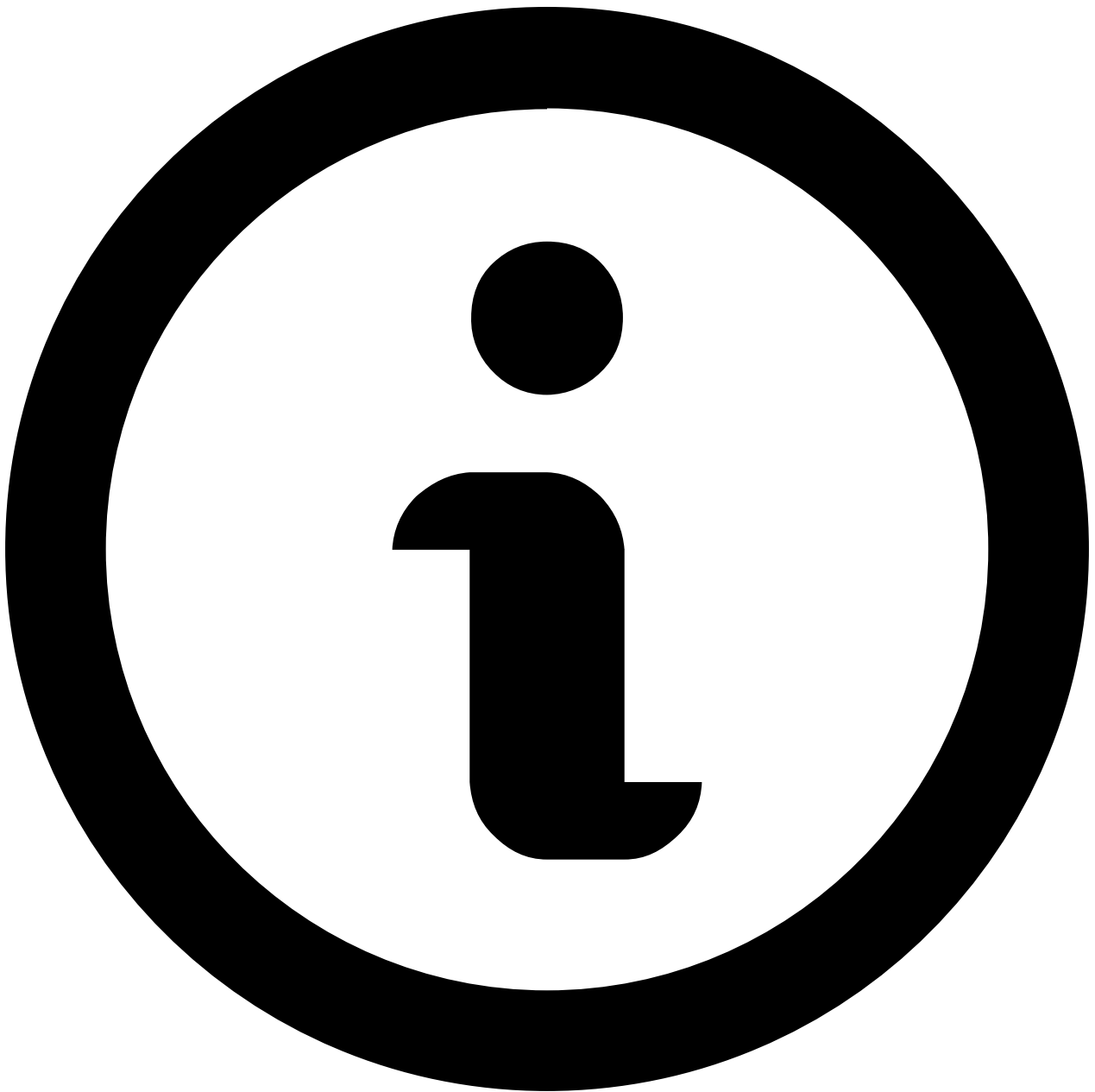
The Workspace Operator role sits between Network Operator and Network Admin. The following table summarizes what you can and cannot do as a Workspace Operator:

Action	Allowed
All Network Operator actions	Yes
Trigger snapshot collection and connectivity tests	Yes
Copy sources from parent network to workspace	Yes
Delete sources	Yes
View credentials, proxies, and jump servers	Yes
Edit and delete snapshots	Yes
Edit and delete networks	Yes
Assign roles to workspace network (up to Workspace Operator)	Yes
Add or edit devices	No
Edit credentials	No
Upload or invalidate snapshots	No

Copying Sources Between Networks^{â}

The required roles for copying sources depend on the type of networks involved:

Source Network	Target Network	Required Role in Source	Required Role in Target
Regular	Regular	Network Admin	Network Admin
Parent	Workspace	Network Operator or higher	Workspace Operator or higher

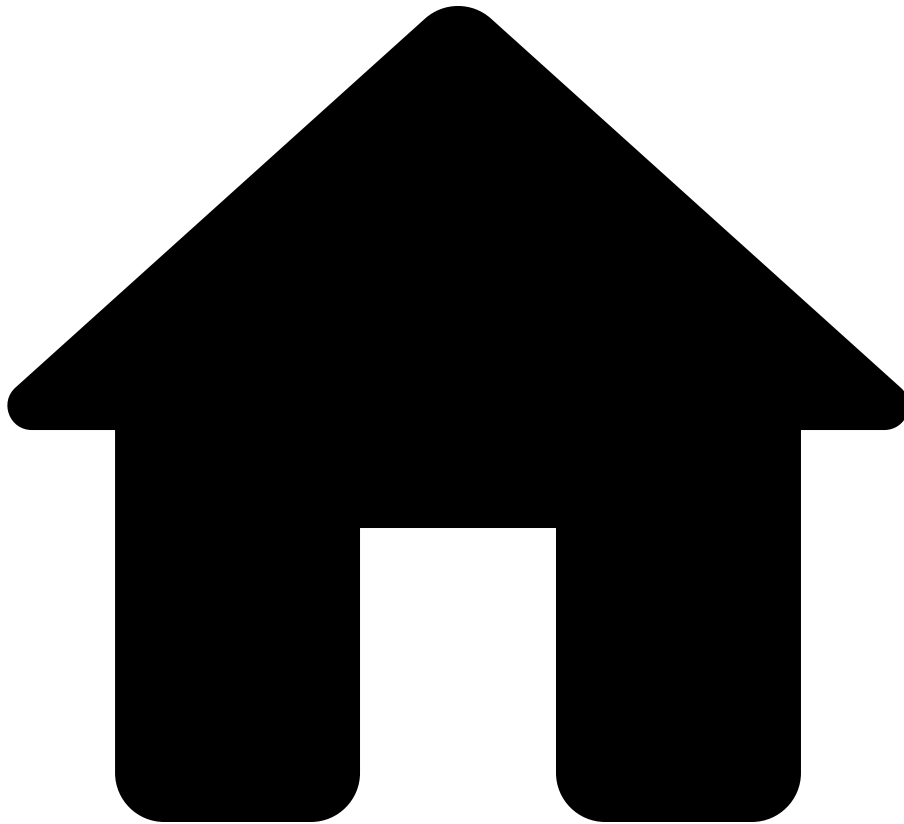


note

As a Network Operator, you can no longer copy sources between two regular (non-workspace) networks. This operation now requires Network Admin in both the source and target networks.

For more details, visit the [Workspace Networks](#) page.

.



- [NQE](#)
- Data Files

On this page

Data Files

Data files let you bring external information—such as site names, asset tags, compliance parameters, or other metadata—into Forward so it can be used in NQE queries, Inventory views, and verification checks. They can include policy definitions, software standards, per-device details, or any other structured dataset your queries rely on.

Data files are defined and stored at an org-level. CSV, JSON, and text formats are supported.

Once uploaded, the data is immediately viewable in NQE.

Data files can also be captured in network snapshots. This can be configured by network operators on a per-network basis.

Access and Permissions[ââ](#)

Who can do what:

- **Org Admins / Network Admins:** Can upload data files.
 - **Network Operators and above:** Can enable or disable collection of data files into the snapshots of a network to which they have access.
-

Managing Data Files[ââ](#)

Data files are uploaded at the **organization level** from **NQE ââ Data Files**, and then individually **enabled or disabled per network** from **Sources ââ Data Files**.

Where to Add (Upload) Data Files[ââ](#)

To upload a new data file, go to: **NQE ââ Data Files**

This is the **only** location where the full import wizard is always available. Use this path whenever you need to:

- upload a new CSV/JSON/TXT file
- replace or update an existing file
- review schema, filename, or NQE name

Steps to upload:

1. Go to **NQE ââ Data Files**.
2. Click **Import data file**.
3. Select your file and review the preview.

Import data file

Upload your data file to begin the import process. Make sure that:

- .json or .csv extensions are in the file name
- STIGs policy file must follow a specific format. Download and fill it out the provided template before uploading it.

[Download template](#) 

Note: Only one STIG policy file can exist in the system at any given time. Uploading a new policy file will replace an older version.

Selected file



sample-data-file.json

File size: 889 B

or

[Change file](#)

Name

sample-data-file.json

NQE name

sample-data-file

Description (optional)

[Cancel](#)

[Next](#)

1. Save to complete the upload.

Review the imported data

Check for any discrepancies between the imported data and recognized structure.

JSON input

Search

Prettify

```
1 {
2   "platforms": [
3     {
4       "vendor": "ARISTA",
5       "series": "7000series",
6       "os": "ARISTA_EOS",
7       "targetVersions": [
8         "4.31.2F"
9       ],
10      "acceptableVersions": [
11        "4.23.15M-2GB",
12        "4.28.7.1M",
13        "4.29",
14        "4.30"
15      ],
16      "retiredVersions": [
17        "4.16",
18        "4.23.13M-2GB",
19        "4.24",
20        "4.25",
21        "4.26.4M"
22      ]
23    },
24    {
25      "vendor": "CISCO",
26      "series": "3560stack",
27      "os": "IOS_XE",
28      "targetVersions": [
29        "17.12"
30      ],
31      "acceptableVersions": [
32        "17.09"
33      ],
34      "retiredVersions": [
35        "17.08"
36      ]
37    }
38  ]
39 }
```

Recognized data structure

Expand all

Collapse all

sample-data-file: Record

platforms: Bag

vendor: String

series: String

os: String

targetVersions: Bag

acceptableVersions: Bag

retiredVersions: Bag

Back

Cancel

Save

After upload, the file becomes available to all networks in the organization via **NQE** 

Data Files

net3

net3

4

?

⚙

noreply2

⏻

Sources

Collection

Health

Platform

NQE

Library

Decorators

Data Files

Tutorial

Collectors

Settings

Forward Docs

Data files

ⓘ

STIG policy file is automatically applied to all networks and can't be modified. To include a non-STIG file in Snapshots, navigate to Sources → Data files

1 data file

4 hidden columns

Import data file

⋮

Name ↓	Description	Data type	Uploaded by	Uploaded	Included in networks	
Filter	Filter	Filter	Filter	Filter	Filter	
sample-data-file.json		JSON file	admin	39 min ago ⓘ	2	📄 🔗 ⬇ ⋮

Enabling or Disabling Data Files per Network⌘

From **Sources ⌘ Data Files**, you manage data files **at the network level**.

Here you can:

- enable or disable data files for that network
- review which files are active within the selected network

net3

net3

4

?

⚙

noreply2

⏻

Snapshot

Search

Inventory

Verify

Security

Diff

Objects

Network

Dashboard

Sources

Collection

Health

Sources

Take Snapshot

Discover sources

<

anaged (0)

VMware NSX & vCenter (2) ▾

Synthetic (1)

Endpoints (2)

Data connectors (0)

Data files (1)

End host scanners (0) >

ⓘ

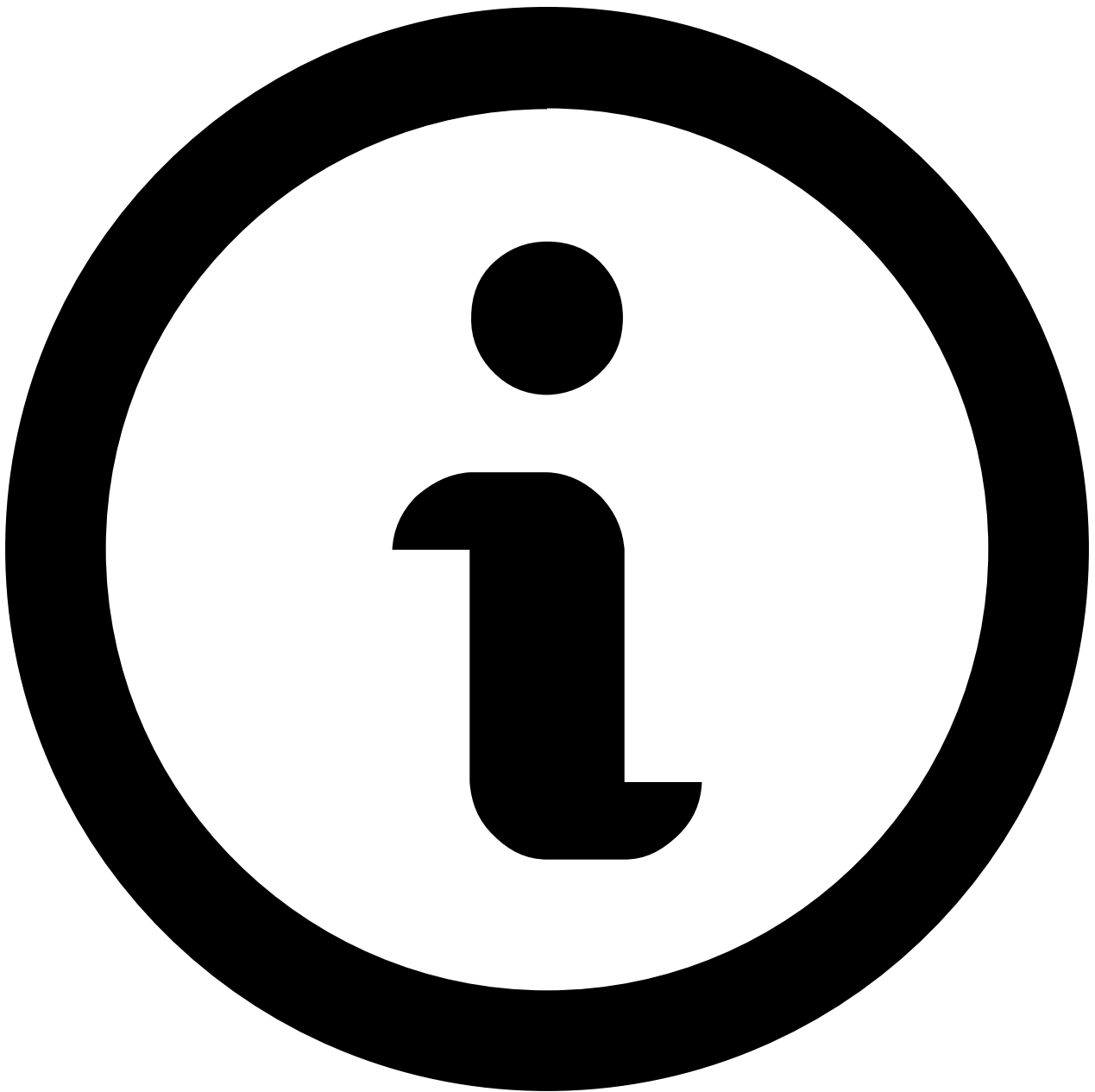
Added data files will apply to future Snapshots

1 data file

4 hidden columns

⋮

Include	Name ↓	Description	Data type	Uploaded by	Uploaded	
Filter ▾	Filter	Filter	Filter	Filter	Filter	
☒	sample-data-file.json		JSON file	admin	34 min ago ⓘ	📄 ⬇



note

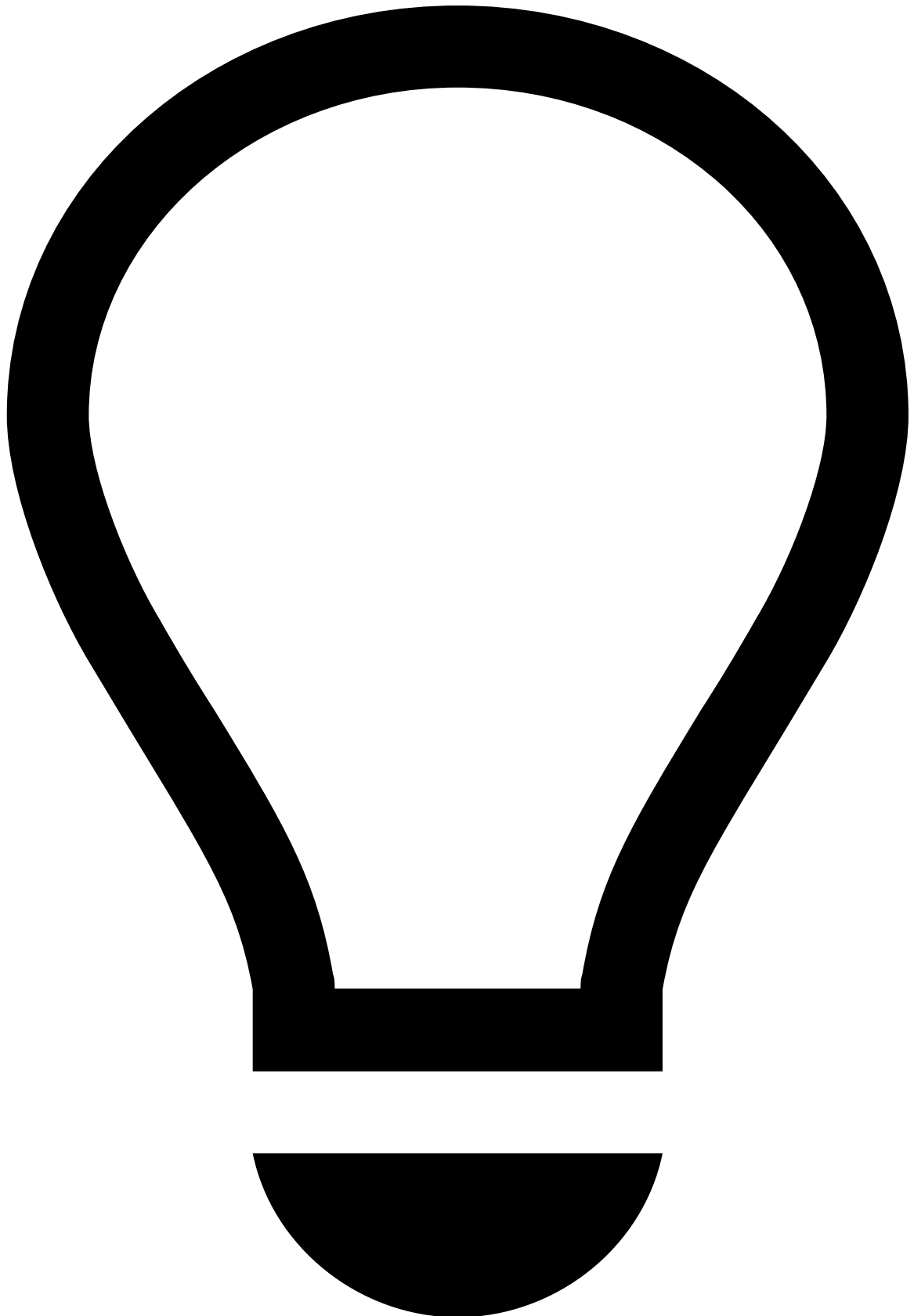
This is *not* the place to regularly add or maintain files.

The **Import** button appears here **only when no data files exist at all** in the organization. Once at least one file exists, this page becomes enable/disableâonly.

Impact on Queries [↗](#)

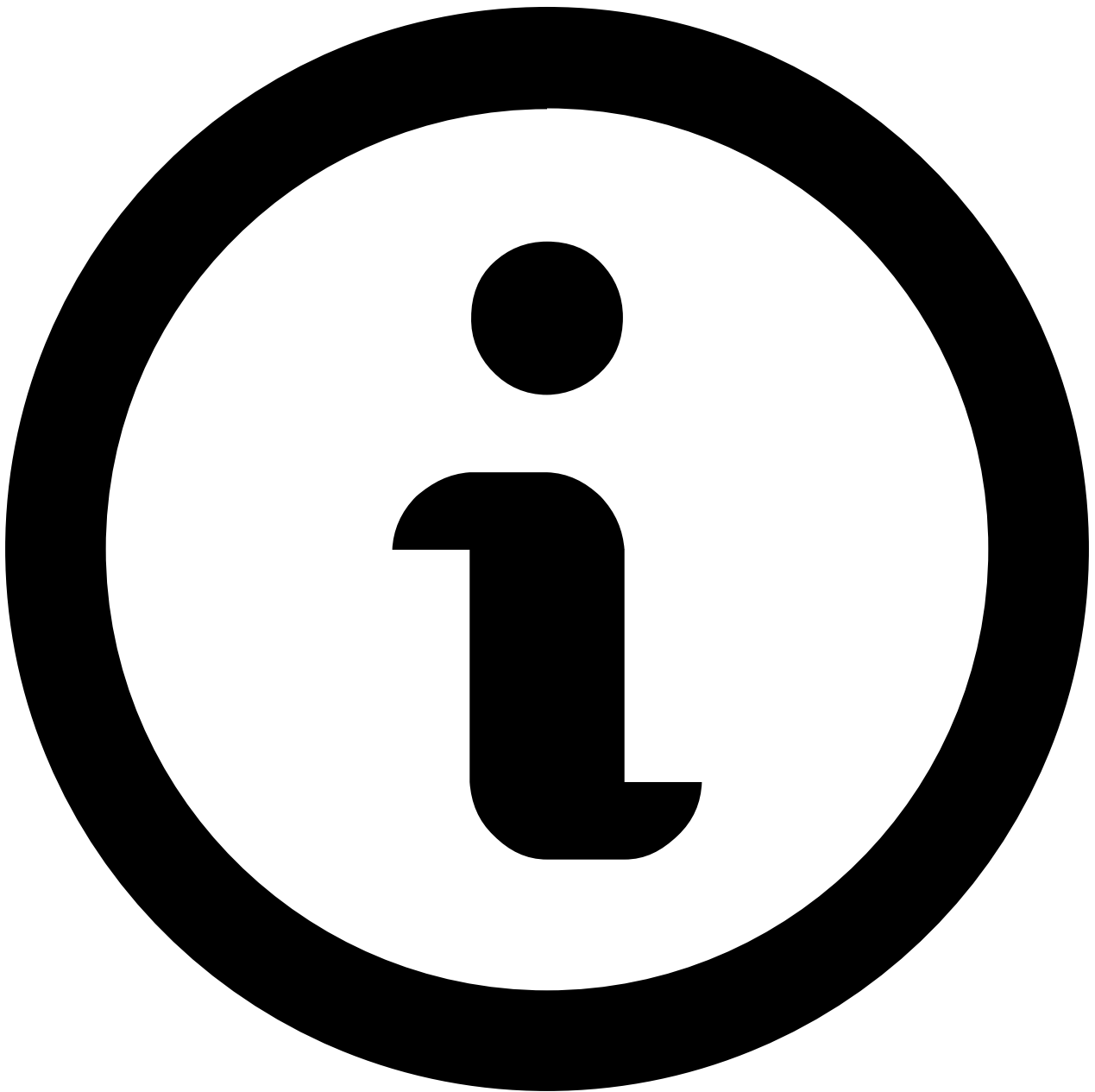
Once a data file is uploaded at the organization level **and enabled for a network**, it is incorporated into all future snapshots and becomes available immediately for use in:

- the NQE data model
- NQE Library queries
- Inventory and Verify views that reference data files



[Important]

Changing a file's *name*, *NQE name*, or *schema* may break existing queries.
Update any queries that reference the file after making changes.



note

If a data file is later disabled, it remains present in previously collected snapshots but will not be included in future snapshots unless re-enabled.

Replacing Data Files [↗](#)

To update an existing data file, click **Replace File**, upload the new version using the same file and NQE names, and confirm the change. If the file's structure differs from the original, the system will warn you before proceeding.

Updating a data file does not modify the contents of any previously collected snapshots. The new version is applied only to future snapshots and to any views that support previewing results with the latest data files.

Running Queries With the Latest Data Files in NQE Library

When a query depends on a data file, the result viewer includes a selector allowing you to run it using:

- **Snapshot** (the authoritative model), or
- **Snapshot with Latest Data Files** (a preview overlay incorporating the newest uploaded file)

This enables immediate validation of data file contents without waiting for the next snapshot.

The screenshot displays the NQE interface. On the left, the 'Software Compliance' tab is active, showing a query editor with a CQL query. The query is as follows:

```
1 platformPolicy =
2   foreach policy in [network.extensions["sample-data-file"]]
3   where isPresent(policy.value)
4   foreach platform in policy.value.platforms
5   select {
6     vendor: fromVendor(platform.vendor),
7     os: fromOS(platform.os),
8     series: platform.series,
9     targetVersions: platform.targetVersions,
10    acceptableVersions: platform.acceptableVersions,
11    retiredVersions: platform.retiredVersions
12  };
13
14 // TODO Extend this to handle any additional vendors we need
15 fromVendor(s) =
16   if s == "ARISTA"
17   then Vendor.ARISTA
```

Below the query editor, the 'Results' section shows a dropdown menu with 'net3' selected. A date selector is open, showing a list of timestamps. The top option is '2025-11-17 08:02:39 UTC' with a checkmark, indicating it is the selected 'Latest Data Files' snapshot. Other timestamps include '2025-11-14 20:02:57 UTC', '2025-11-14 18:51:01 UTC', '2025-11-14 18:26:58 UTC', and '2025-11-14 18:19:29 UTC'. An 'Execute' button is visible next to the date selector.

On the right, the 'Data Model' tab is active, showing a hierarchical tree structure. The tree includes nodes for 'network: Network', 'devices: Bag<Device>', 'endpoints: Bag<NetworkEndpoint>', 'locations: Bag<Location>', 'cloudAccounts: Bag<CloudAccount>', 'cloudOrganizationalUnits: Bag<OrganizationalUnit>', 'externalSources: ExternalSources', 'dataConnectors: Bag<DataConnector>', 'cveDatabase: CveDatabase', 'stigDatabase: StigDatabase', and 'extensions: NetworkExtensions'. The 'sample-data-file' node is expanded, showing its status as 'DataFileStatus' and its value as a bag of platforms.

Behavior notes:

- The selector only appears for queries that reference data files.
- If no data files used by the query have changed since the snapshot, the **Latest Data Files** option does not appear.

- Running with latest data files does **not** modify the underlying snapshot.
-

Using Latest Data Files in Inventory+[â](#)

Inventory+ also supports the ["latest data files"](#) preview for queries that depend on them. See the [Inventory+](#) for more information.

Using Latest Data Files in Verify[â](#)

Verification pages (including STIG checks) support a **read-only preview** that applies the latest data files on top of the snapshot.

Key points:

- Verification **status** always reflects the snapshot.
 - The latest-data-files mode is only a preview of how results would look in the next snapshot.
 - The toggle appears only for checks that depend on data files.
-

Data Files for STIG Security Policies[â](#)

Parameterized STIG checks use operator-supplied policy files, uploaded through the Data Files interface.

To begin:

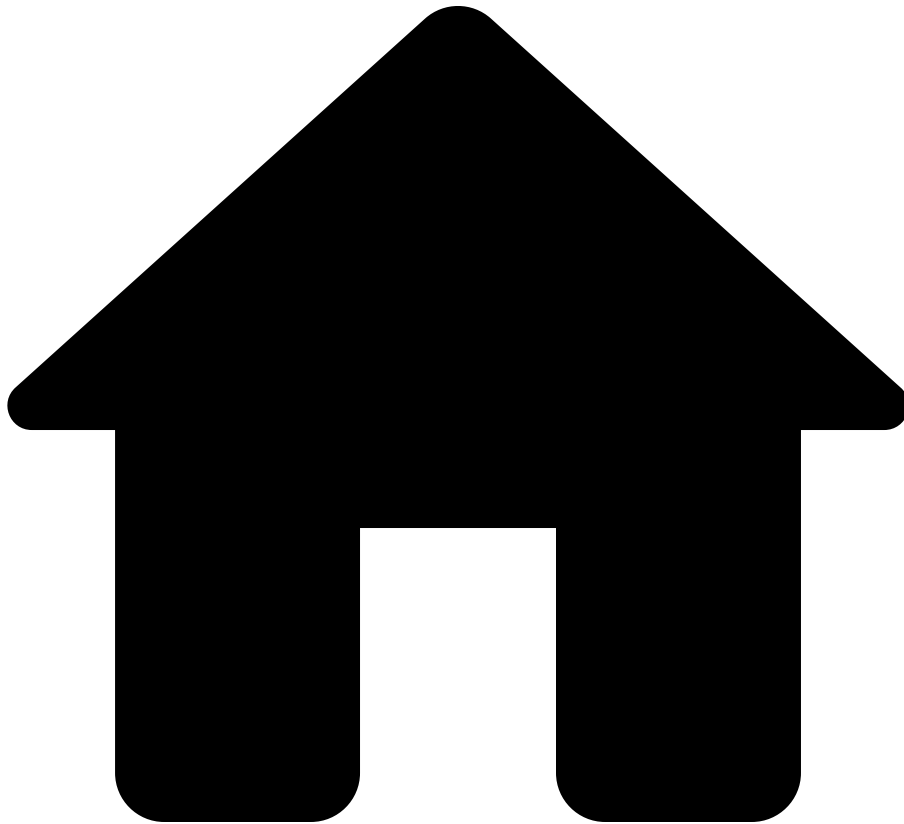
1. Go to **Data Files** [â](#) **Import data file**.
2. Select **Download Template** to obtain the expected CSV structure.
3. Fill in the device, interface, or rule parameters.
4. Upload the completed file.

After upload, you can Preview STIG results immediately using ["snapshot with latest data files"](#).

Limitations

- The published NQE API returns only **snapshot-based** results; it does not expose the latest-data-files preview.
- Diffs app does not compare snapshot vs snapshot-with-latest-data-files.
- Snapshot data remains authoritative; latest-data-files mode is a preview overlay only.

.



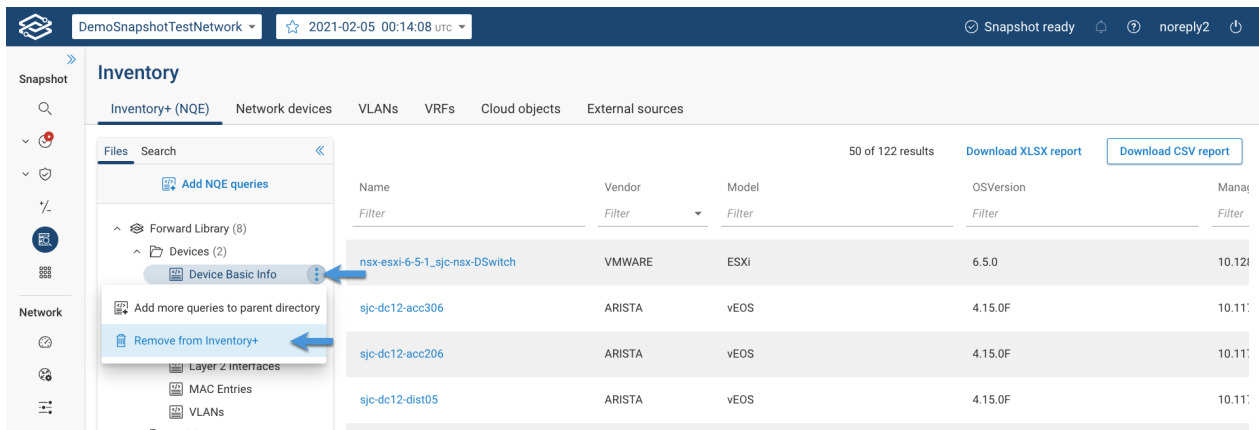
- [Inventory](#)
- Inventory+ (NQE)

On this page

Inventory+ (NQE)

Inventory+ provides seamless access to network data across various technological domains, including detailed insights into device hardware and Layer 2 (L2) and Layer 3 (L3) configurations and states.

Inventory+ utilizes Forward's [Network Query Engine \(NQE\)](#), presenting data views built from NQE queries available in the [NQE Library](#).



The queries in Inventory+ are processed each time a new Snapshot is taken, ensuring that the latest data is always accessible. Results are displayed in a tabular format, which can be sorted and filtered as needed.

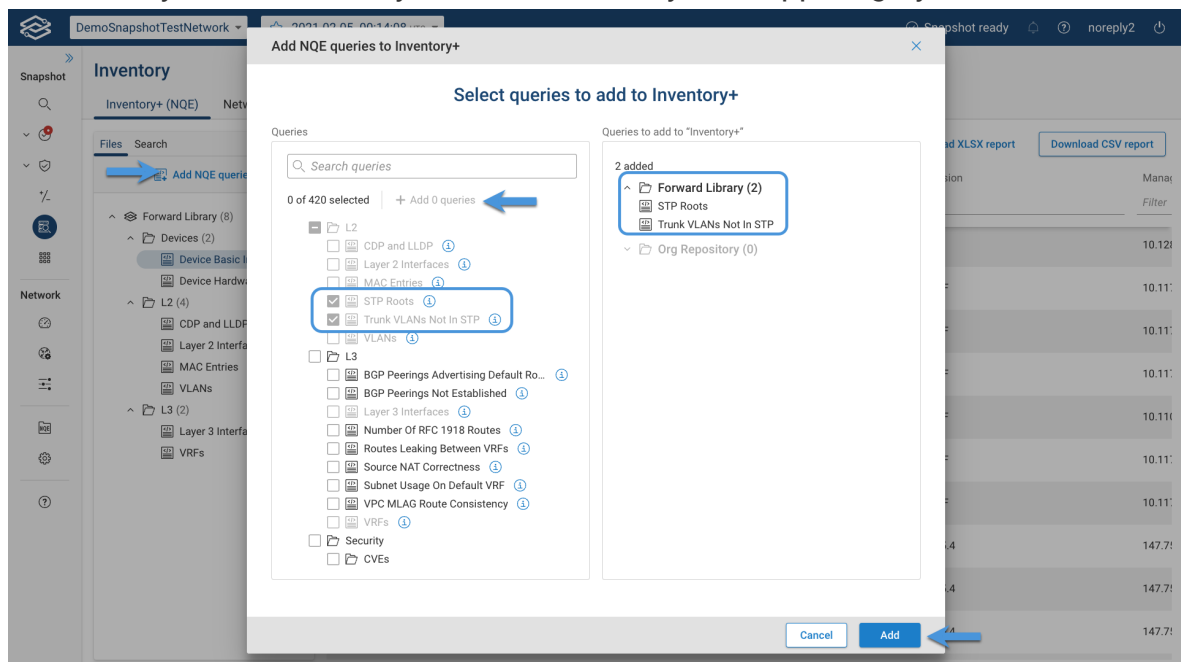
Add Queries to Inventory+ [🔗](#)

Inventory+ allows users to add queries in two ways:

- [Using the Add NQE queries button](#)
- [From a folder](#)

Using the Add NQE Queries Button [🔗](#)

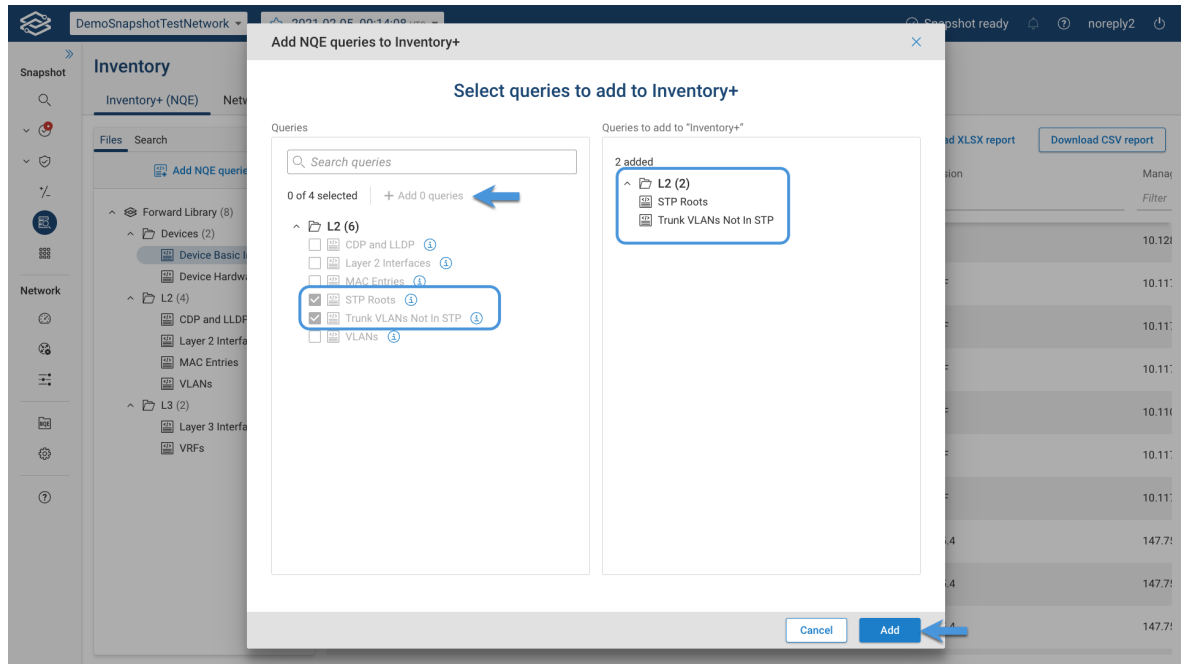
1. Click the **Add NQE queries** button located above the query list.
2. A dialog box will appear with the complete list of available NQE queries from the NQE Library. Queries already added to Inventory+ will appear greyed out.



3. Select queries to add, either individually or using the bulk selection option.
4. Click **Add x queries** and confirm by clicking **Add**.

From a Folder

1. Open the folder's menu by clicking the **kebab icon** (three vertical dots) next to the folder name.
2. A dialog box will appear, showing the list of queries within the selected folder. Queries already present in Inventory+ will appear greyed out.



3. Select the desired queries, click **Add x queries**, and confirm by clicking **Add**.

Remove Queries from Inventory+

Removing Individual Queries

1. Locate the query in the list.

- Click the **kebab icon** (three vertical dots) next to it and select **Remove from Inventory+**.

The screenshot shows the 'Inventory' section of a network management tool. The left sidebar has a 'Files' panel with a search bar and a list of folders: 'Forward Library (8)', 'Devices (2)', 'Layer 2 Interfaces', 'MAC Entries', and 'VLANs'. The 'Devices (2)' folder is expanded, showing a list of devices. A context menu is open for the device 'nsx-esxi-6-5-1_sjc-nsx-DSwitch', with the 'Remove from Inventory+' option highlighted. The main table displays a list of devices with columns: Name, Vendor, Model, OSVersion, and Manager. The table shows 50 of 122 results.

Name	Vendor	Model	OSVersion	Manager
nsx-esxi-6-5-1_sjc-nsx-DSwitch	VMWARE	ESXi	6.5.0	10.121
sjc-dc12-acc306	ARISTA	vEOS	4.15.0F	10.111
sjc-dc12-acc206	ARISTA	vEOS	4.15.0F	10.111
sjc-dc12-dist05	ARISTA	vEOS	4.15.0F	10.111

Removing Folders

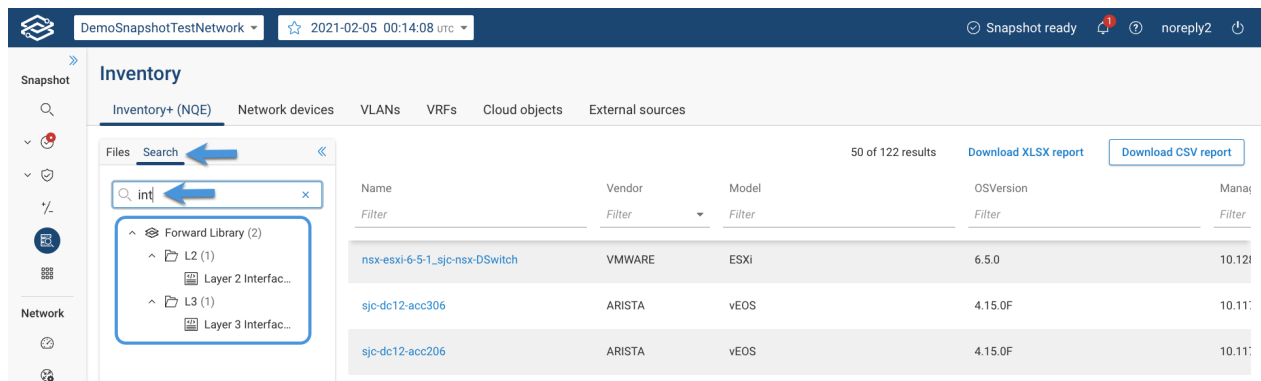
- Locate the folder in the list.
- Click the **kebab icon** (three vertical dots) next to it and select **Remove x queries from Inventory+** to remove the folder and all its queries.

The screenshot shows the 'Inventory' section of a network management tool. The left sidebar has a 'Files' panel with a search bar and a list of folders: 'Forward Library (8)', 'Devices (2)', 'L2 (4)', 'L3 (2)', 'VLANs', 'Layer 3 Interfaces', and 'VRFs'. The 'L2 (4)' folder is expanded, showing a list of interfaces. A context menu is open for the interface 'atl-ce01', with the 'Remove 4 queries from Inventory+' option highlighted. The main table displays a list of interfaces with columns: Device, Interface, Neighbor, and Neighborface. The table shows 50 of 714 results.

Device	Interface	Neighbor	Neighborface
atl-ce01	ge-0/0/2	atl-dc01-spine02	et16
atl-ce01	ge-0/0/3	atl-dc01-spine01	et16
atl-ce01	ge-0/0/6	atl-core-pe01	516
atl-ce01	ge-0/0/8	atl-ce02	521
atl-ce01	ge-0/0/9	atl-core-pe02	516
atl-ce01	ge-0/0/10	atl-isp-edge01	518

Searching Queries

The **Search** tab enables users to find queries by matching their summaries to the search text.



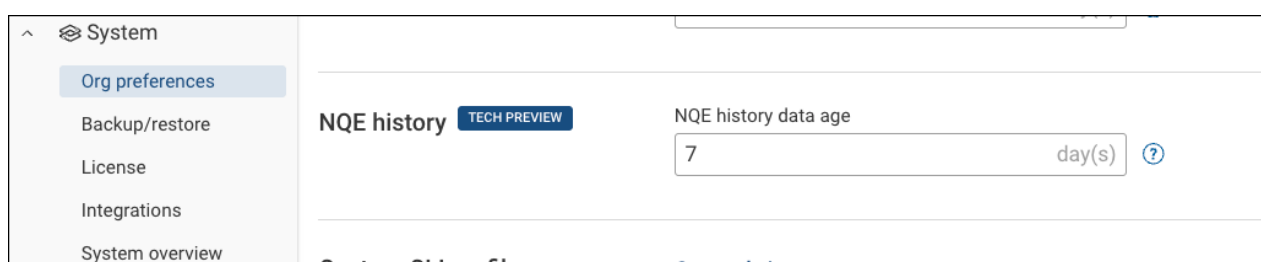
NQE History

NQE History enables users to view how network data changes across multiple snapshots and identify when selected data, such as a device, an interface, or an IP address, changed without manually inspecting each snapshot. It provides historical visibility for selected NQE queries in **Inventory+**, recording when selected key attributes changed across processed snapshots and displaying that information alongside the most recent snapshot results.

NQE History records and retains data from snapshots that have been processed since history tracking was enabled for an NQE query. Historical results remain available until they expire based on the configured retention period (more on this below).

Key features of NQE History:

- Any snapshot processed since history tracking was enabled will contribute data to the historical view.
- Results from each snapshot are retained until they expire.
- The retention period is configured in **Settings > System > Org Preferences > NQE History** (default: 7 days with a maximum value of 30 days).



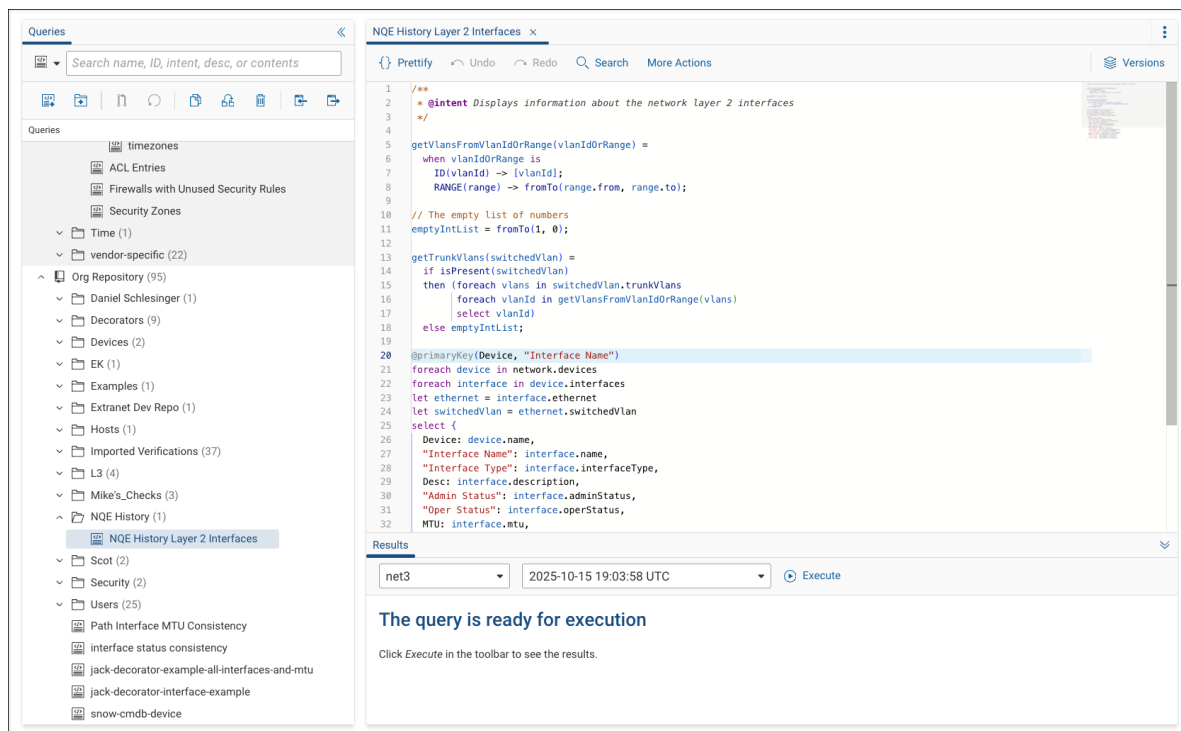
Enabling NQE History🔗🔗

To enable historical tracking, an NQE query must define one or more **primary keys** that identify unique records within the results.

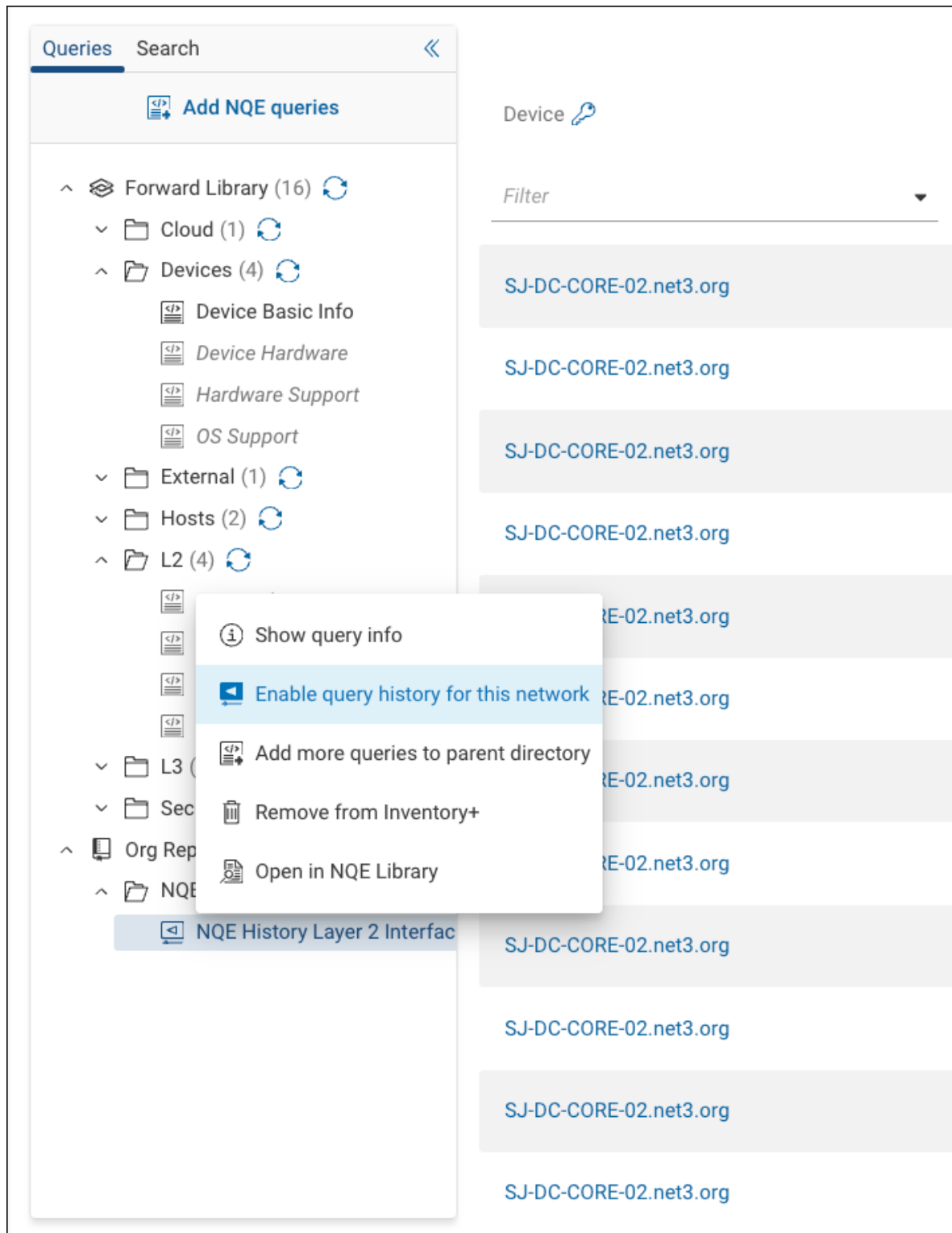
For example, if you need to track how the IP addresses changes in your network you would create an NQE query where you would âprimary keyâ the IP address, its subnet and its source so that the NQE query will generate an entry for each unique triplet and track each of them for the configured retention period.

1. In the **NQE Library**, create or open a query.
2. Add a primary key annotation to specify which attributes uniquely identify each record.

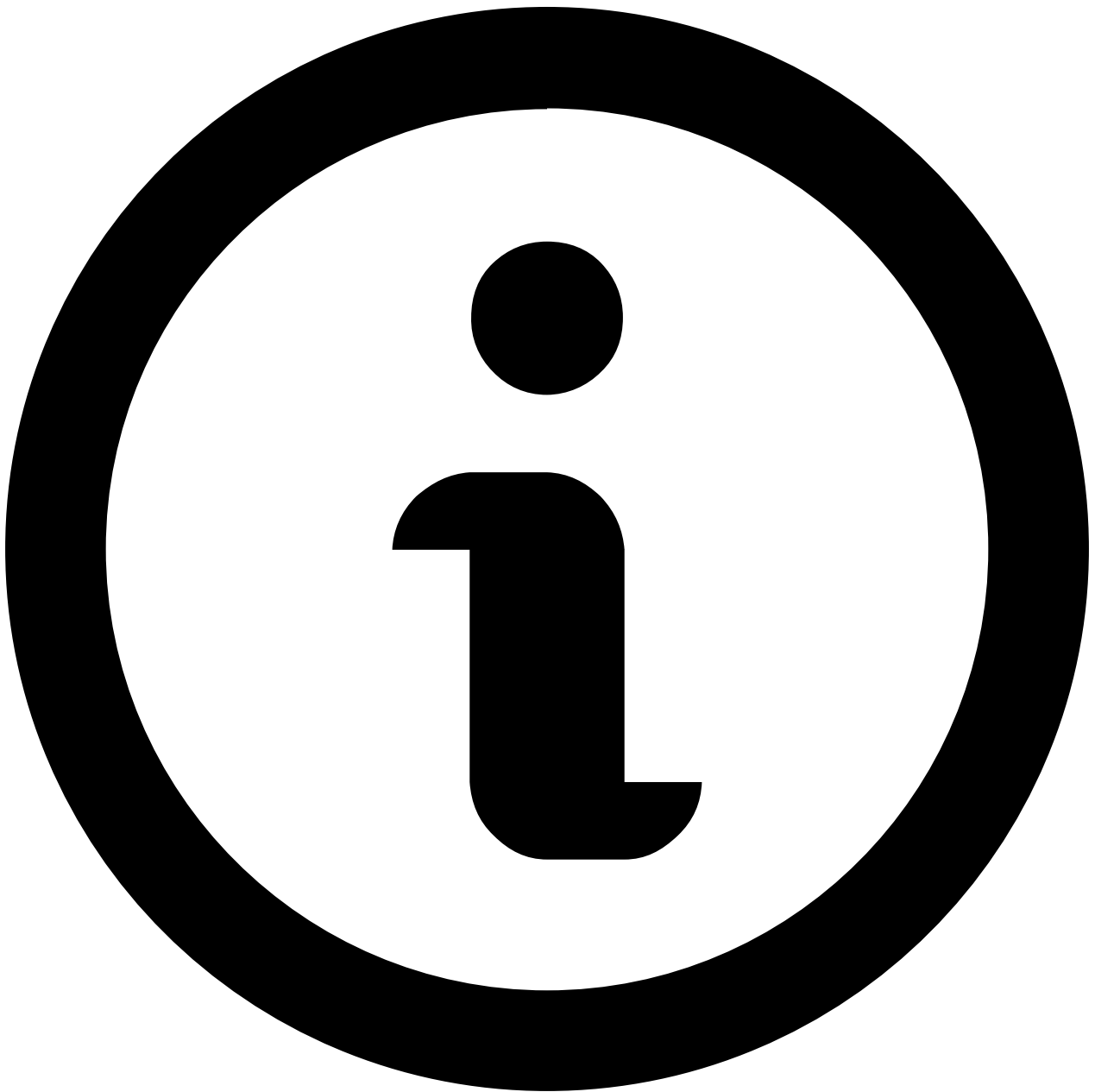
```
@primaryKey(Device, "Interface Name")
```



3. Commit the query changes.
4. In **Inventory+**, add the NQE query. Once added, right-click the query and select **Enable query history for this network**.
 - Queries that support history are marked with a **historical** icon next to the query name.



5. After enabling the NQE query, the system collects historical data from all processed snapshots within the configured retention period.



note

Editing or removing a query with history enabled

- Editing a query clears the stored history and repopulates it using currently processed snapshots.
- Removing a query clears all stored history and does not repopulate it.
- If a query is edited in a way that makes it ineligible for history, NQE History is disabled for that query across all networks and its historical data is deleted. A

query becomes ineligible if it is made invalid, if the primary key is removed, or if columns named **Last Seen** or **Historical Status** are added.

Per-network configuration

History tracking is configured per network. Enabling NQE History for a query in one network does not automatically enable it for the same query in other networks.

Disabling NQE History [🔗](#)

To disable NQE History for a query, right-click the query in **Inventory+** and select **Disable query history for this network**.

Queries

Search

⏪

📄 Add NQE queries

^ 📁 Forward Library (16) 🔄

📁 Cloud (1) 🔄

^ 📁 Devices (4) 🔄

📄 Device Basic Info

📄 Device Hardware

📄 Hardware Support

📄 OS Support

📁 External (1) 🔄

📁 Hosts (2) 🔄

^ 📁 L2 (4) 🔄

📄

📄

📄

📄

📁 L3 (

📁 Sec

^ 📁 Org Rep

^ 📁 NQE

📄 NQE History Layer 2 Interfac

snapshot

historical

Includes data up to 5

Device 🔑

sj-dc-vcenter10_sj-dc-esxi-144.net3.or... × ▾

sj-dc-vcenter10_sj-dc-esxi-144.net3.org_DSv ...

sj-dc-vcenter10_sj-dc-esxi-144.net3.org_DSv ...

sj-dc-vcenter10_sj-dc-esxi-144.net3.org_DSv ...

er10_sj-dc-esxi-144.net3.org_DSv ...

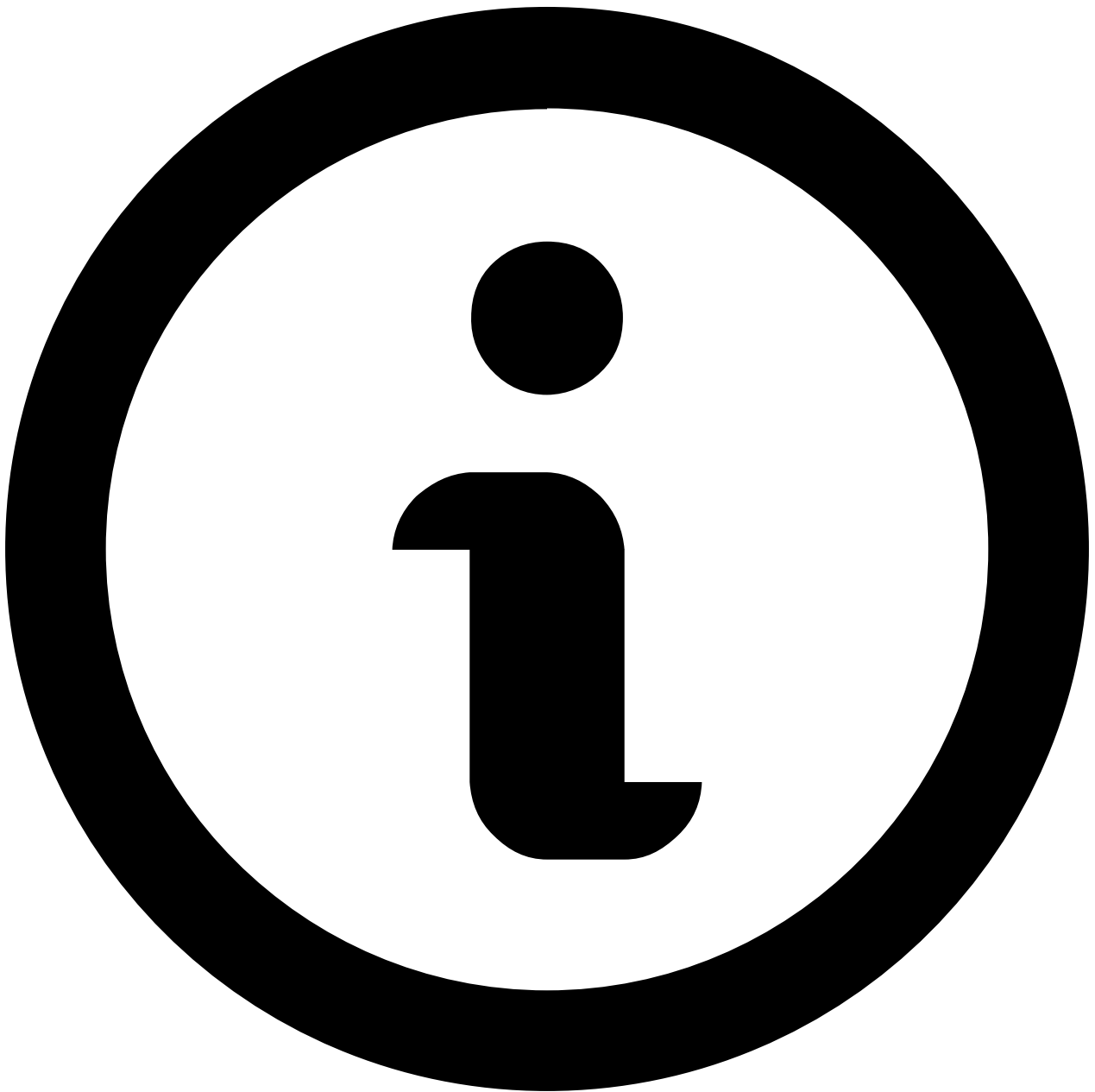
🔍 Show query info

ⓧ Disable query history for this network

📄 Add more queries to parent directory

🗑 Remove from Inventory+

📄 Open in NQE Library



note

Disabling history removes all stored historical data for the query.

Viewing Historical Data [↗](#)

When a query has NQE History enabled, **Inventory+** displays two tabs:

- **Snapshot:** Shows the data for the most recent snapshot.
- **Historical:** Displays data from all processed snapshots within the retention period.

Inventory

Inventory+ (NQE) Network devices Cloud objects Endpoints

Queries Search snapshot **historical** Includes data up to 5 days ago ⓘ

4 of 4 filtered results 9 hidden columns Download report

Device	Interface Name	Interface Type	Admin Status	Oper Status	MAC Address	Last seen	Historical status
sj-dc-vcen10_sj-dc-esxi-144.net3.org...	Filter	Filter	Filter	Filter	Filter	Filter	Filter
sj-dc-vcen10_sj-dc-esxi-144.net3.org_DSv ...	144-linux-11_nic1	IF_ETHERNET	UP	UP	00:50:56:a8:52:fe	2025-10-15 07:01:11 UTC	stale
sj-dc-vcen10_sj-dc-esxi-144.net3.org_DSv ...	144-linux-6_nic1	IF_ETHERNET	UP	UP	00:50:56:a8:06:fa	2025-10-15 07:01:11 UTC	stale
sj-dc-vcen10_sj-dc-esxi-144.net3.org_DSv ...	144-linux-8_nic1	IF_ETHERNET	UP	UP	00:50:56:a8:8a:e1	2025-10-15 07:01:11 UTC	stale
sj-dc-vcen10_sj-dc-esxi-144.net3.org_DSv ...	vmnic0	IF_ETHERNET	UP	UP	50:01:00:2c:00:00	2025-10-15 19:03:58 UTC	fresh

The historical view includes additional columns:

- **Last Seen:** The most recent snapshot in which the record appeared.
- **Historical Status:** Indicates whether the record is *Fresh* (currently present) or *Stale* (no longer present in the current snapshot).

Columns representing the query's primary key are marked with a **key icon**.

Using Latest Data Files in Inventory+ [↗](#)

Inventory+ queries that rely on data files include a selector allowing you to choose whether to view:

- snapshot-based results, or
- snapshot **with the latest data files**

This selector appears only when the underlying query uses a data file and a new version of that file has been uploaded.

See [NQE Data Files](#) for more details on working with data files..

Inventory

Inventory+ (NQE) Network devices Cloud objects Endpoints

Queries Search

Add NQE queries

Forward Library (17)

- Cloud (1)
- Devices (4)
- External (1)
- Hosts (2)
- L2 (4)
- L3 (3)
- Security (2)
 - STIGs (1)
 - STIG Findings
 - ACL Entries

Historical Snapshot Latest data files

500 of 9,358 results 0 hidden columns Download report

violation	Device	Vulnerab... ID	Rule ID	Rule Version
Filter	Filter	Filter	Filter	Filter
Indeterminate	SJ-DC-CORE-02.net3.org	V-217005	SV-217005r945858_rule	CISC-RT-000310
Pass	SJ-DC-CORE-02.net3.org	V-216741	SV-216741r531087_rule	CISC-RT-000070
Pass	SJ-DC-CORE-02.net3.org	V-216523	SV-216523r991929_rule	CISC-ND-000140
Pass	SJ-DC-CORE-02.net3.org	V-216531	SV-216531r960993_rule	CISC-ND-000530

End-of-Life/End-of-Support Data

Inventory+ includes prebuilt queries (OS Support and Hardware Support) that directly consume the lifecycle data from NQE data model.

See [NQE HW/SW Lifecycle](#) for more details.

Downloading a Report

Inventory+ allows users to export query results in various formats for analysis or compliance purposes. Available export formats include **CSV**, **XLSX**, and **CKLB**.

Downloading a CSV and XLSX Report

1. Open **Inventory+** and select your desired query.
2. In the results pane, click on **Download report** at the top-right corner.
3. Select one of the following options from the dropdown:
 - **Download .csv report**
 - **Download .xlsx report**

- The selected report will download to your system. Any active sorting or filters are preserved in the exported file.

The screenshot shows the Forward Networks Inventory+ (NQE) interface. The top navigation bar includes the Forward Networks logo, a date/time selector (2024-10-29 18:05:04 UTC), and a user profile (admin). The left sidebar contains a 'Snapshot' section with 'Inventory' selected, and a 'Network' section with 'Dashboard', 'Sources', 'Collection', and 'NQE' options. The main content area is titled 'Inventory' and shows a list of 'STIG Findings'. The table has columns for 'violation', 'Device', and 'Vulnerability ID'. The results show five entries, all with a 'Fail' status and the device 'bigip3k1'. A dropdown menu is open on the right, showing options to 'Download .xlsx report', 'Download .csv report', and 'Download .ckl report (STIG)'. The 'Download .ckl report (STIG)' option is highlighted.

violation	Device	Vulnerability ID
Fail	bigip3k1	V-217383
Fail	bigip3k1	V-217384
Fail	bigip3k1	V-217385
Fail	bigip3k1	V-217386
Fail	bigip3k1	V-217387

CKLB Export

CKLB Export generates STIG compliance reports in a DoD-compliant [CKLB format](#). This export option is available only for queries that include all the required columns found in any of the **STIG Findings** queries (e.g., STIG Findings - Cat I, STIG Findings - Cat II, or STIG Findings - Cat III). While additional columns can be added to the query, none of the required columns can be removed.

Steps to download a CKLB report:

- Execute any of the **STIG Findings** queries within Inventory+.
- In the results pane, click on **Download report** at the top-right corner.
- Select **Download .ckl report (STIG)** from the dropdown menu.
- The selected report will download to your system in .ckl format for use with DoD compliance tools like STIG Viewer.

Forward Networks2024-10-29 18:05:04 UTCadmin

Snapshot

Search

Inventory

Verify

NQE

Predefined

Intent

Security

Posture

Blast Radius

Vulnerability

Exposure

Diff

Objects

Network

Dashboard

Sources

Collection

NQE

1 item

Inventory

Inventory+ (NQE)Network devicesCloud objectsCustom sources

QueriesSearch

Add NQE queries

Forward Library (15)

Cloud (1)

Cloud Accounts

Devices (3)

Device Basic Info

Device Hardware

OS Support

External (1)

Hosts (2)

L2 (4)

L3 (3)

Security (1)

STIGs (1)

STIG Findings

500 of 1,192 results0 hidden columnsDownload report

violationDeviceVulnerability ID

FilterFilterFilter

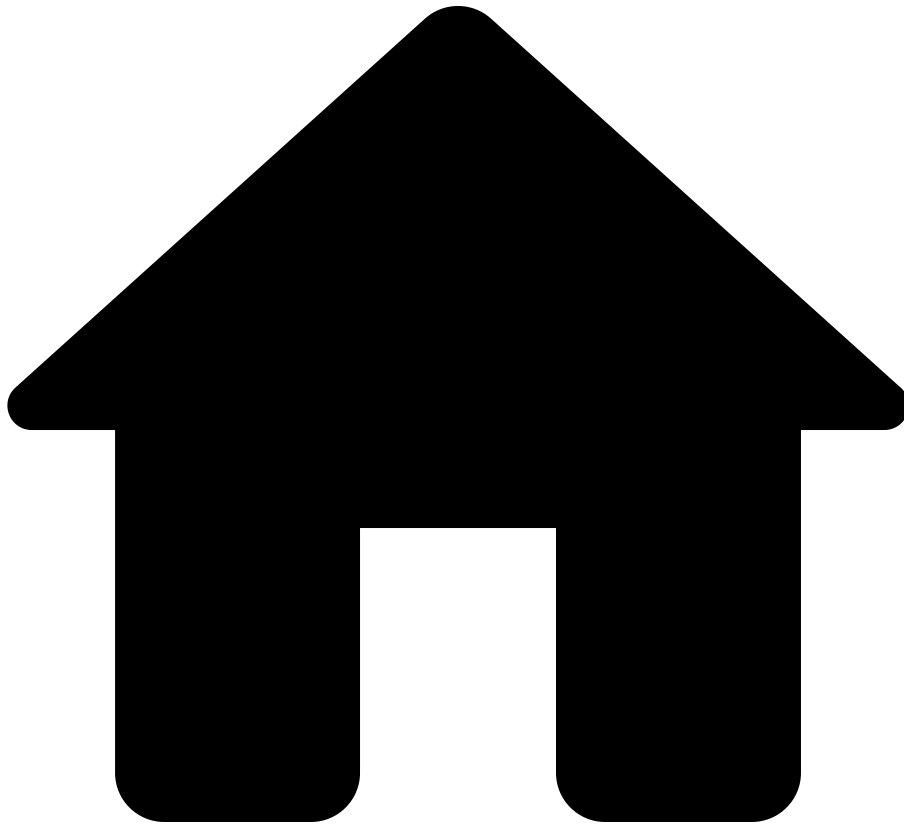
Fail	bigip3k1	V-217383	SV-217383r879525_rule
Fail	bigip3k1	V-217384	SV-217384r879526_rule
Fail	bigip3k1	V-217385	SV-217385r879527_rule
Fail	bigip3k1	V-217386	SV-217386r879528_rule
Fail	bigip3k1	V-217387	SV-217387r879530_rule

Download .xlsx report

Download .csv report

Download .csl report (STIG)

.



- [Sources](#)
- [Configure Collection](#)
- Data Connectors

On this page

Data Connectors

Data connectors integrate and augment network data from sources external to the network itself. They enable Forward Enterprise to retrieve, parse, and structure data using **HTTP(S) GET requests**. Data connectors can enhance visibility, improve analytics, and integrate external data sources into Forward Enterprise.

Navigating the Data Connectors page

You can view and manage data connectors by navigating to the **Sources** page and selecting the **Data connectors** category.

Sources

Take Snapshot

Discover sources

Categories

«

Q Search for IP, device, type, tags, k

Network devices

Classic devices (89)

Cloud-managed (1)

Cloud providers (0)

VM platforms

vCenters (1)

NSX-T (1)

Synthetic nodes (3)

Endpoints (2)

Data connectors (1)

Data files (2)

End host scanners (0)

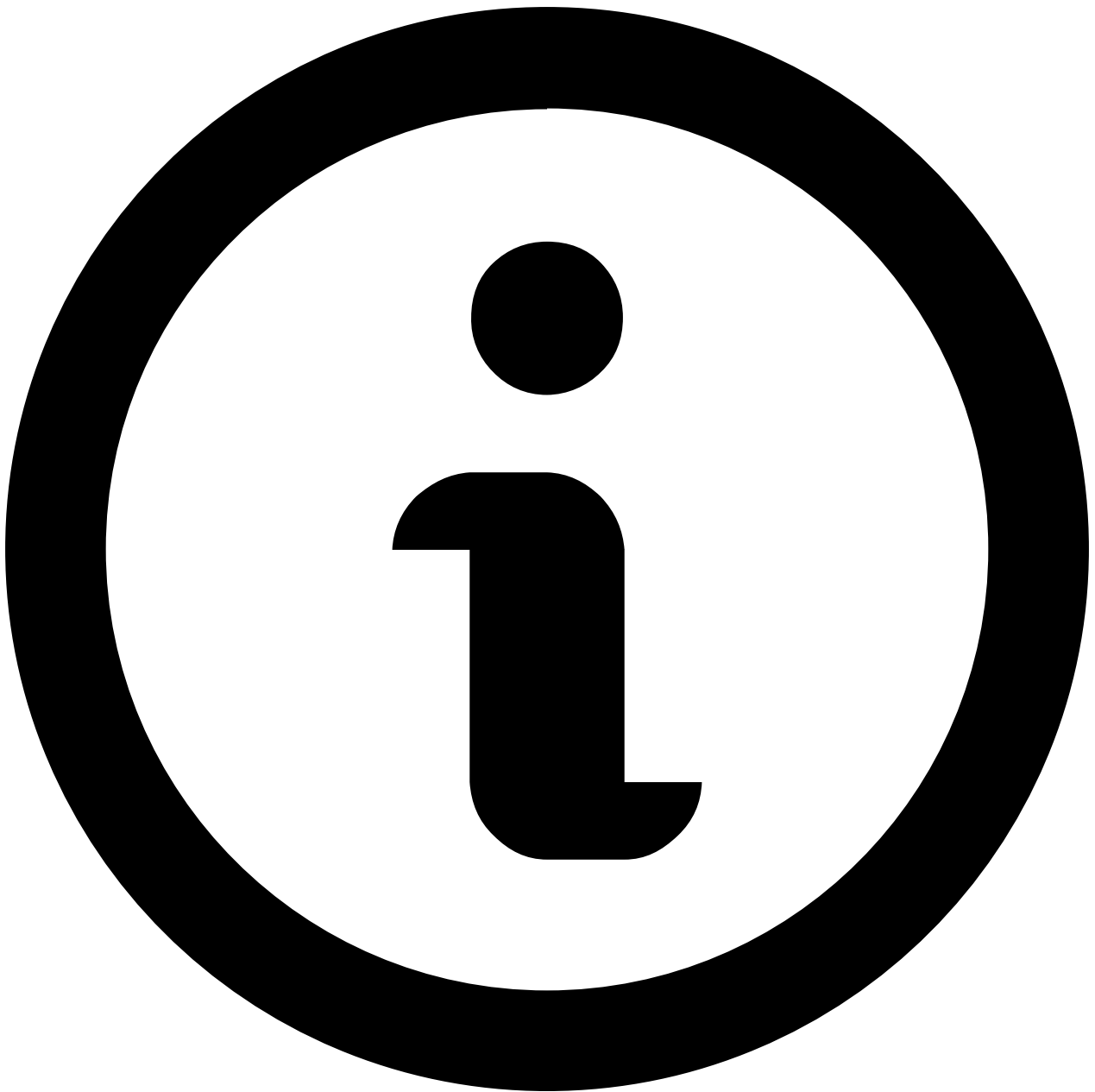
1 source | 4 hidden columns | + Add data connector | ⋮

Collect	Name ↓	IP/host	Type	Credentials	Connectivity test status	Latest Snapshot status ?	
Filter	Filter	Filter	Filter	Filter	Filter	Filter	
☒	rest-data-lake	api.restful-api.dev	HTTPS	none	connected	not collected	↺ ↻ 🗑

1. **Add Data Connector:** Launches the **Add HTTP(S) Data Connector** wizard.
2. **Additional Options (⌵® Kebab Menu):** Dropdown with option to **reset column width**.
3. **List of Data Connectors:** Displays a filterable list of data connectors with the following columns:
 - **Collect:** Toggle to enable or disable data collection.
 - **Name:** Displays the assigned name.
 - **Tags:** Assigned tags for categorization.
 - **IP/host:** The IP address or hostname of the data connector.
 - **Type:** The profile type assigned to the data connector.
 - **Credentials:** Credentials used for authentication.
 - **Connection Test Status:** Status of the last connection test.
 - **Latest Snapshot Status:** Status of the most recent snapshot.
 - **Last Tested:** Timestamp of the last connection test.
4. **Per-Connector Actions:** Data connector-specific actions available at the end of each row:
 - **Test connection:** Initiates a real-time connectivity test for the selected data connector.
 - **Edit:** Edit the selected data connector configuration.
 - **Delete:** Deletes the data connector.

Adding an HTTP(S) Data Connector [â](#)

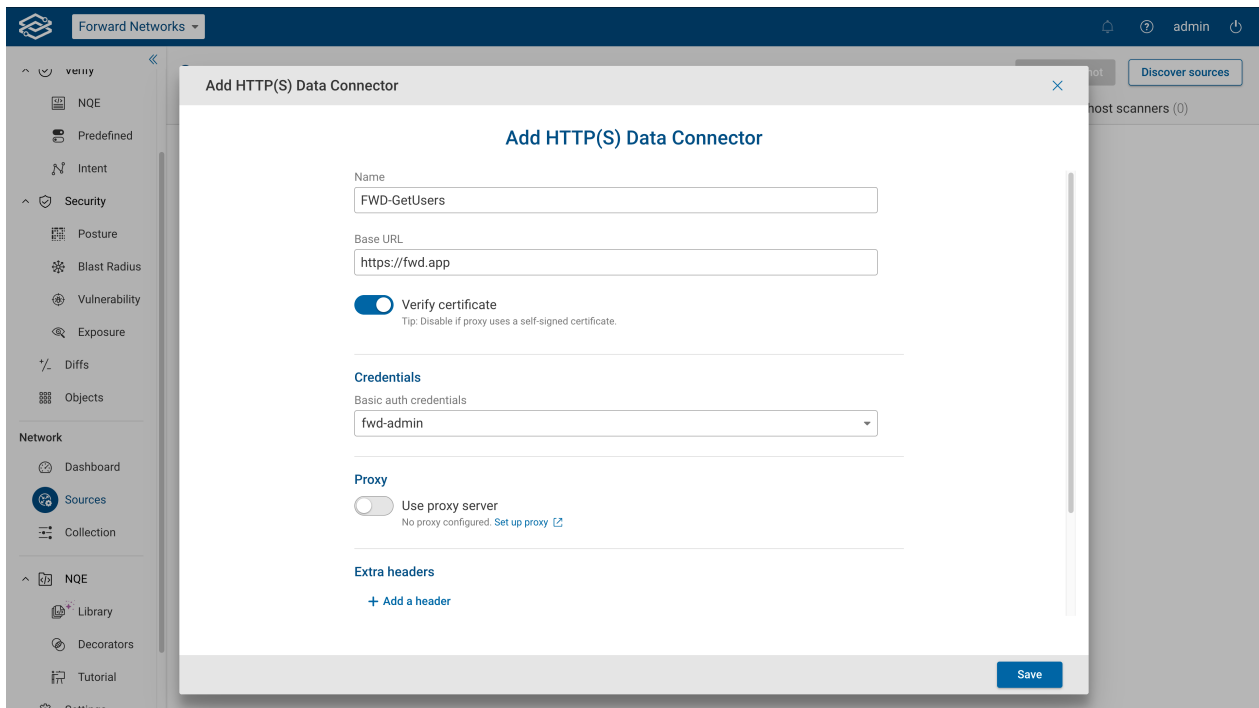
To add a data connector, select **Add data connector** to launch the wizard.



note

Forward supports data in JSON, CSV, and any text. Binary data is not supported.

The format is based on the `content-type` header. If the content type is set to `application/json` then Forward uses JSON format. If the content type is set to `text/csv` then Forward will use CSV format; otherwise, it uses text.



From the **Add HTTP(S) Data Connector** wizard, provide the following details:

General Information

- **Name:** Assign a name for the data connector.
- **Base URL:** Specify the base URL for requests.

Credentials

- **Basic auth credentials:** Select or create credentials from the dropdown.
- **Use proxy server:** Enable if a proxy is required. Select or create one from the list.

Extra headers

- **Add extra headers:** Define additional headers as **Key/Value** pairs.

Endpoints

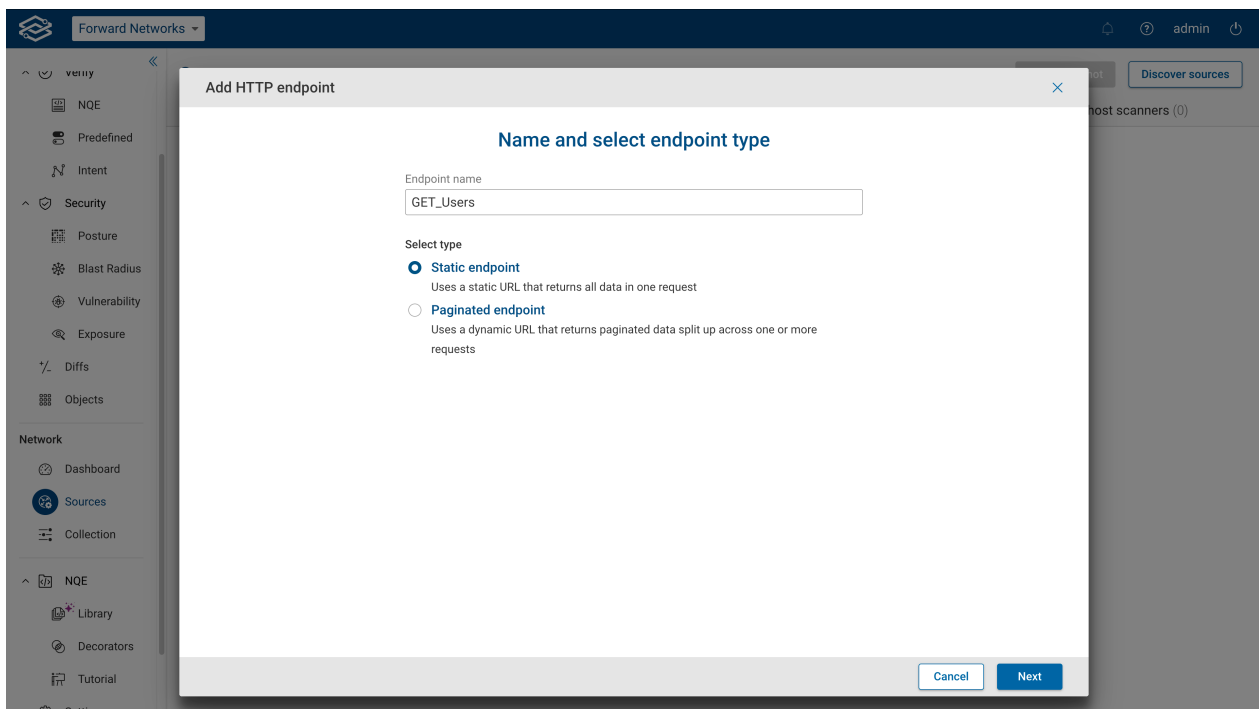
- **Add an endpoint:** Select to assign endpoints to the data connector. There are two types to select from:
 - [Static Endpoints](#): Uses a static URL that returns all data in one request.

- [Paginated Endpoints](#): Uses dynamic URLs for paginated data split up across one or more requests.

Once all endpoints have been added to the data connector, click **Save**.

Adding a static endpoint

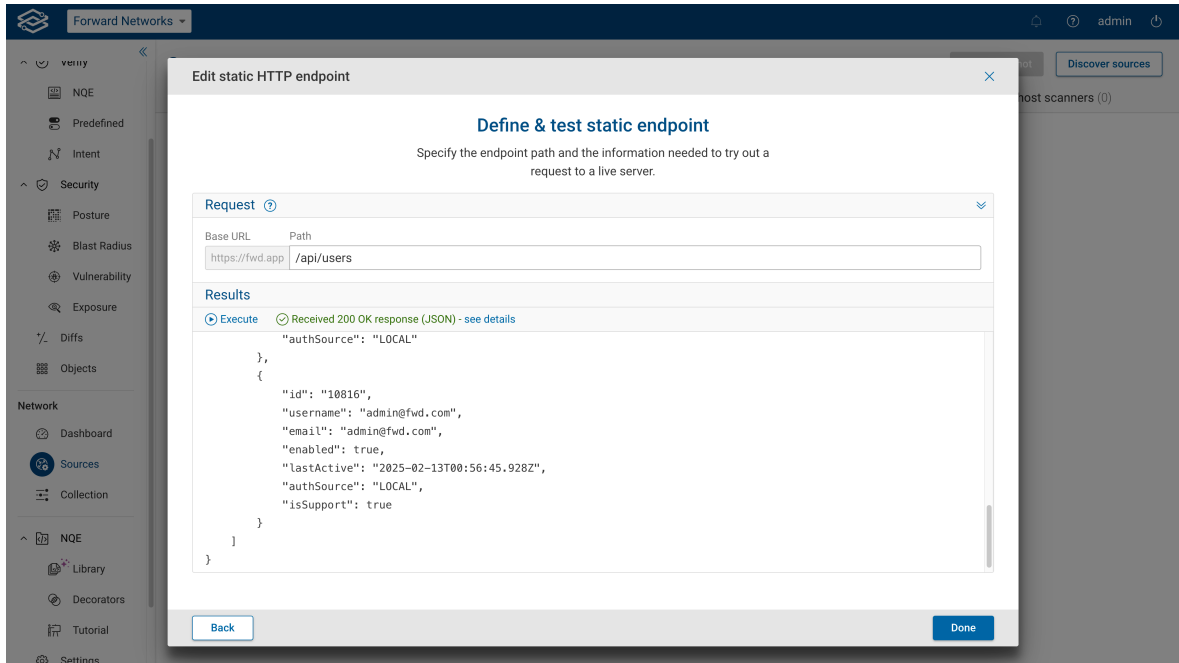
A **static endpoint** uses a fixed URL that returns all data in a single request. This is useful for retrieving datasets that do not require multiple calls or pagination to be fully collected.



To configure **static endpoint**:

1. Provide a **name**, and select **static** as the endpoint type.
2. Enter the **request path** and click **Execute** to test connectivity. Results of the connectivity test will be displayed within the **Results** section. Additional

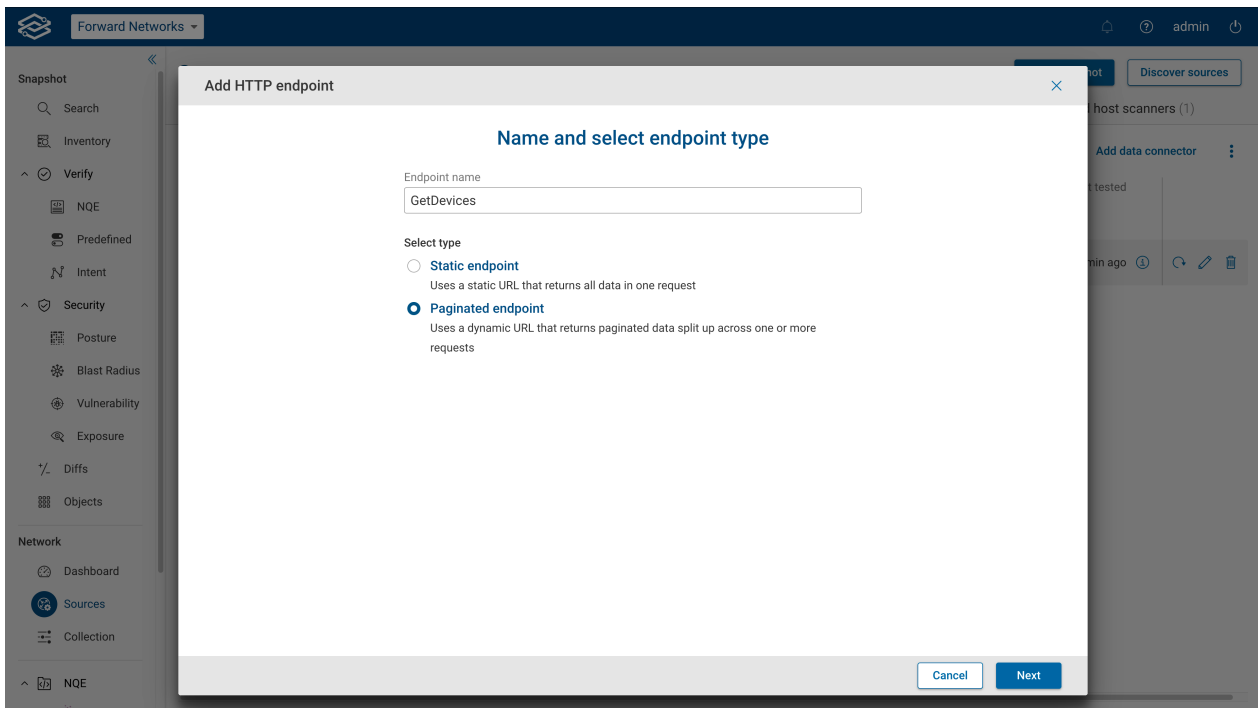
information can be found by selecting **see details**.



3. Once the test succeeds, select **Save**.

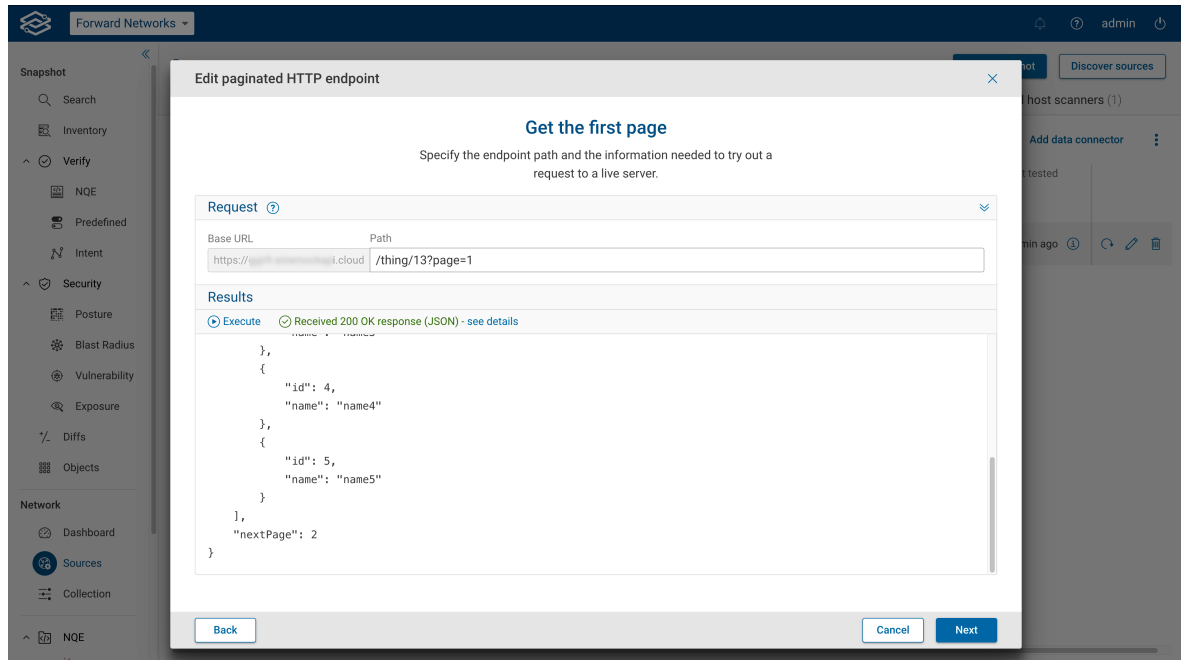
Adding a paginated endpoint

A **paginated endpoint** is used when data is split across multiple pages, requiring multiple requests to retrieve the full dataset. Paginated data sources help manage large volumes of data efficiently.



To configure **paginated endpoint**:

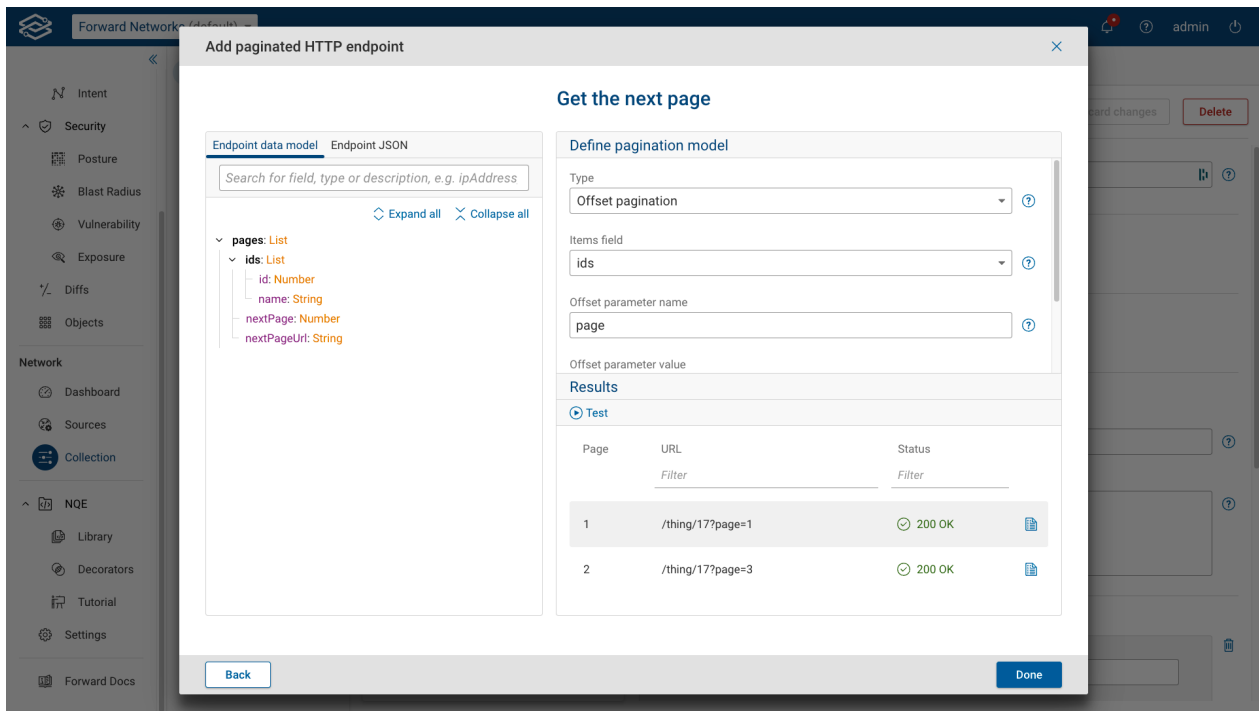
1. Provide a **name** and select **paginated** as the endpoint type.
2. Enter the **request path** for the first page and click **Execute** to test connectivity.
Results of the connectivity test will be displayed within the **Results** section.
Additional information can be found by selecting **see details**.



3. Once the test succeeds, select **Next**.
4. Choose the pagination model **type** for getting the next page:
 - **Offset pagination**: Uses an offset and limit query parameters.
 - **Parameter cursor pagination**: Uses a parameter value in responses.
5. Define the pagination parameters, test the setup, and click **Save**.

Offset pagination

Offset pagination is a method where the request specifies an offset index and a limit to determine the number of results returned in each request. This approach is commonly used in APIs where data is stored in sequential order.

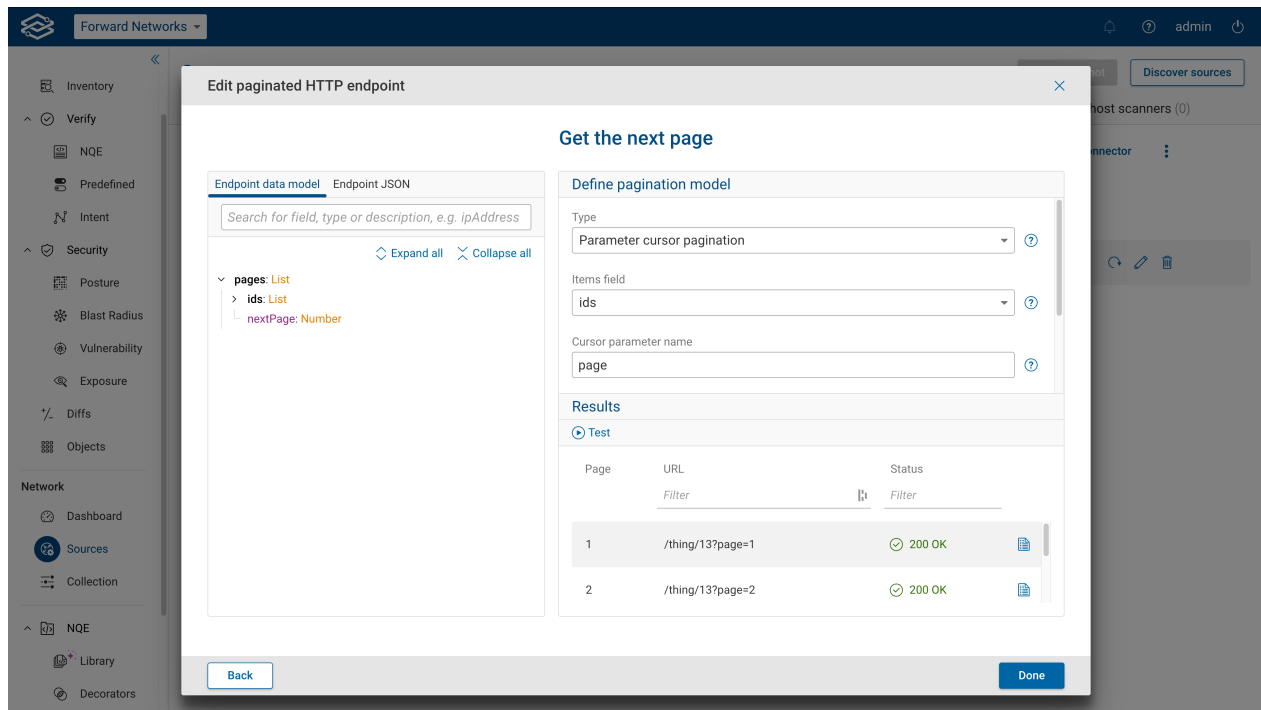


To configure **offset-based pagination**:

1. Select **Offset-based pagination** as the **pagination model type**, and provide the following:
 - **Items Field**: Field in the response data model that contains the paginated items.
 - **Offset Parameter Name**: Name of the offset parameter in query-string.
 - **Offset Parameter Value**: Define how the **offset parameter value** should be incremented for each subsequent page request.
 - **Previous Offset + Length**: This option uses the value of the selected **Items field** to calculate the next page.
 - **Previous Offset + nextPage**: This option uses the value of the **nextPage** in the response to calculate the next page.
 - **Previous Offset + 1**: This option uses the value of **1** to calculate the next page.
 - **Max Number of Pages** (optional): Define the maximum number of pages to be collected.
2. **Test** the setup and click **Done** if successful. Additional information for each result is available by selecting the **see details** icon located at the end of each row.
3. Select **Done** after completing a successful test to finish.

Parameter cursor pagination

Parameter cursor pagination uses a dynamically generated cursor value provided in the response to fetch the next set of data. This method is more efficient than offset pagination, as it avoids skipping records when data changes dynamically.



To configure **parameter cursor pagination**:

1. Select **Parameter cursor pagination** as the **pagination model type**, and provide the following:
 - **Items field**: Field in the response data model that contains the paginated items.
 - **Cursor parameter name**: Name of the cursor parameter in query-string.
 - **Cursor parameter field**: Specify the field from the response that includes the cursor or token.
 - **Cursor parameter value**: Define how the **offset parameter value** should be incremented for each subsequent page request.
 - **Max number of pages** (optional): Define the maximum number of pages to be collected.
2. **Test** the setup and click **Done** if successful. Additional information for each result is available by selecting the **see details** icon located at the end of each row.
3. Select **Done** after completing a successful test to finish.

Managing data connectors [↗](#)

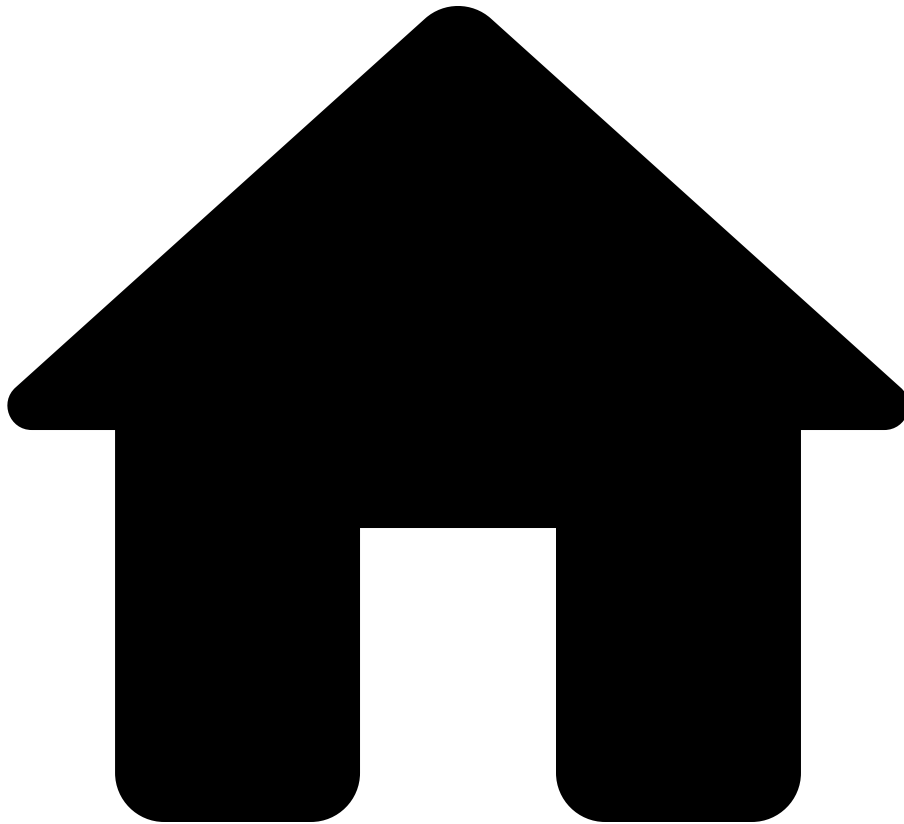
Editing a data connector [↗](#)

To edit a connector, select the **edit** icon in the corresponding row.

Deleting a data connector [↗](#)

To delete a data connector select the **delete** in the corresponding row and confirm the deletion. Please note: deleting a data connector is irreversible.

•

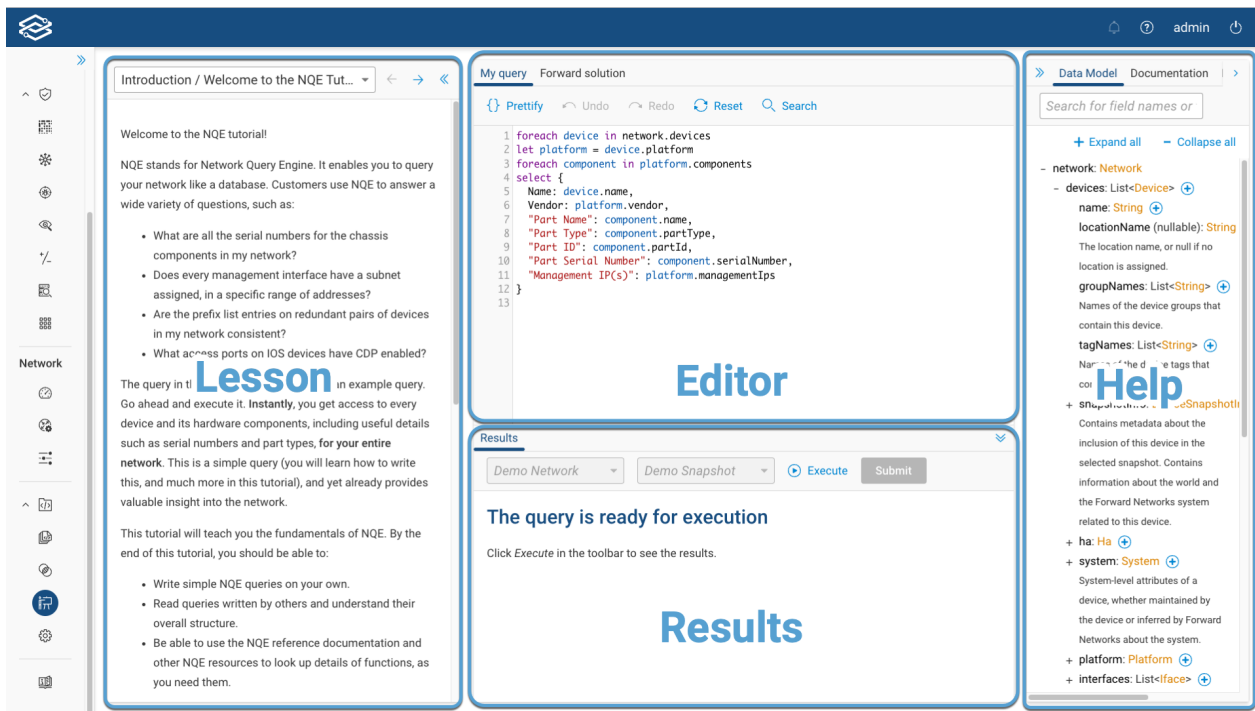


- [NQE](#)
- Tutorial

Tutorial

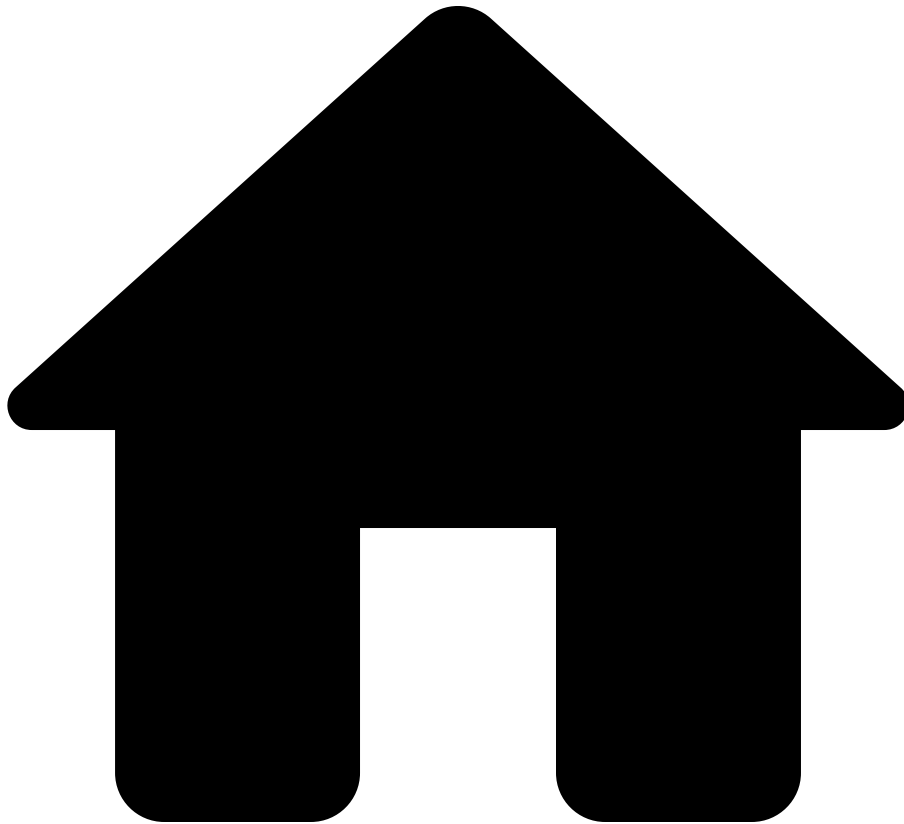
The **NQE Tutorial** is a comprehensive guide consisting of a series of self-contained lessons. Its purpose is to instruct users on the fundamental principles and progressively guide them through the more advanced aspects of NQE (Network Query Engine). Navigate to **NQE > Tutorial** to get started.

The NQE Tutorial is presented in a structured layout with four distinct panes:



- **Lesson** - Located at the top of the **Lesson Pane** is the **Lesson Navigator**. Selecting the dropdown provides a list of all lessons available within the NQE Tutorial, along with their respective progress indicators. Below the Lesson Navigator you will find instructions related to the current lesson.
- **Editor** - The **Editor Pane** is used to create and edit NQE queries. Your work within the editor is saved for each lesson, enabling you to pick up where you left off even after logging out. All content within the editor are private and accessible only to you.
- **Results** - Use the **Results Pane** to see what happens when you run your NQE query. Select **Execute** to run a query and display the results in the pane. When satisfied with the results, select **Submit** to have the tutorial review and grade your query. Successfully completing of a lesson depends on meeting all the goals of the lesson challenge. If issues are identified, the tutorial will provide feedback on ways to improve.
- **Help** - The **Help Pane** serves as a reference for additional information and assistance. Use it to explore the **NQE Data Model**, refer to **Documentation**, and examine useful **Examples**.

.



- [NQE](#)
- HW/SW Lifecycle

On this page

HW/SW Lifecycle

Forward's Network Query Engine (NQE) includes structured **End-of-Life (EoL)** and **End-of-Support (EoS)** information for both operating systems and hardware components across supported network platforms. This data is surfaced through the NQE data model under each device's `platform.osSupport` and `platform.components.support` fields.

This page provides an overview of the purpose, behavior, and scope of this EoL/EoS data.

Why This Matters

Tracking lifecycle milestones across diverse vendor platforms is typically manual and error-prone. Embedding lifecycle metadata directly into the network model ensures that Forward can:

- Identify devices that are running software or hardware versions that are unsupported or approaching end-of-support.
 - Provide consistent lifecycle information across vendors
 - Support automated, snapshot-aware analysis
 - Eliminate the need for operators to manually maintain or reference vendor lifecycle sheets
-

What the NQE Data Model Provides

Operating System Lifecycle (platform.osSupport)

For each recognized OS family, the model includes:

- `network > devices > platform` : Normalized OS identifiers
- `network > devices > platform > osVersion` : Version ranges covered
- `network > devices > platform > osSupport > lastMaintenanceDate` : End of maintenance
- `network > devices > platform > osSupport > lastVulnerabilityDate` : End of vulnerability support
- `network > devices > platform > osSupport > lastSupportDate` : End of complete support
- `network > devices > platform > osSupport > announcementUrl` : References to vendor lifecycle documentation

Devices whose OS does not match any entry in the data model will simply not have lifecycle data returned.

Hardware Component Lifecycle (platform.components.support)[^](#)[v](#)

For hardware, the model covers Part-level lifecycle milestones (if published by vendor):

- `network > devices > platform > components > partId` : part identifiers
- `network > devices > platform > components > support > lastMaintenanceDate` : End of maintenance
- `network > devices > platform > components > support > lastVulnerabilityDate` : End of vulnerability support
- `network > devices > platform > components > support > lastSupportDate` : End of complete support
- `network > devices > platform > components > partType` : Structured attributes such as part type, version, serial-number applicability, etc.
- `network > devices > platform > components > support > announcementUrl` : Vendor reference URLs

Hardware components not defined in the data model will not be associated with EoL/EoS information.

Integration with Inventory+ (NQE)[^](#)[v](#)

Inventory+ includes prebuilt queries (**OS Support** and **Hardware Support**) that directly consume this same lifecycle data from NQE. These queries allow users to view EoL/EoS information immediately—no code or custom queries required—using the data defined in the NQE model. Only platforms included in the lifecycle dataset appear in these results; devices running unsupported OSes or hardware simply do not show up in the OS/Hardware Support views. The library queries provide a convenient, UI-driven way to explore lifecycle status across all supported devices in a snapshot.

OS Support

Prettify

Undo

Redo

Search

NQE Assist

More Actions

Versions

```
12 foreach device in network.devices
13 let platform = device.platform
14 let osSupport = platform.osSupport
15 where isPresent(osSupport)
16 select {
17   Device: device.name,
18   Vendor: platform.vendor,
19   Model: platform.model,
20   OS: platform.os,
21   "OS Version": platform.osVersion,
22   "End of OS maintenance": osSupport.lastMaintenanceDate,
23   "End of OS vulnerability": osSupport.lastVulnerabilityDate,
24   "End of OS support": osSupport.lastSupportDate,
25   URL: osSupport.announcementUrl,
26   Location: device.locationName,
27   Tags: device.tagNames,
28   "Collection IP": device.snapshotInfo.collectionIp,
29   "Management IP(s)": platform.managementIps,
30   Type: platform.deviceType
31 }
```

Results

net3.org

2025-11-24 12:08:20 GMT-08:00

Execute

45 of 45 results

Device	Vend...	Model	OS	OS Versi...	End of OS maintena...	End of OS vulnerabi...	End of OS support	URL
Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
SJ-DC-RM10-ACC ...	ARISTA	vEOS	ARISTA_EOS	4.25.2F	2023-10-20	2023-10-20	2023-10-20	https://ww
SJ-DC-RM10-ACC ...	ARISTA	vEOS	ARISTA_EOS	4.25.2F	2023-10-20	2023-10-20	2023-10-20	https://ww
SJ-DC-S-NGFW-A	PALO_ALI ...	PA-VM	PAN_OS	9.0.4	2022-03-01	2022-03-01	2022-03-01	https://ww
SJ-DC-RM10-NGF ...	FORTINET	FortiGal ...	FORTINET	7.4.3	2026-05-11	2027-11-11	2027-11-11	https://enc
SJ-DC-RM17-BL-€ ...	ARISTA	vEOS-lab	ARISTA_EOS	4.31.2F	2026-10-13	2026-10-13	2026-10-13	https://ww
SJ-DC-RM17-LEA ...	ARISTA	vEOS	ARISTA_EOS	4.25.2F	2023-10-20	2023-10-20	2023-10-20	https://ww
SJ-DC-RM15-SPI# ...	ARISTA	vEOS	ARISTA_EOS	4.25.2F	2023-10-20	2023-10-20	2023-10-20	https://ww

Data Model

Generate

Explain

Documentation

Search for field, type or description, e.g. ipAddress, JUMP_SERVER, *

Expand all

Collapse all

network: Network

devices: Bag<Device>

name: String

locationName (nullable): String

groupNames: Bag<String>

tagNames: Bag<String>

accessLabelNames: Bag<String>

snapshotInfo: DeviceSnapshotInfo

ha: Ha

system: System

platform: Platform

deviceType: DeviceType

vendor: Vendor

model (nullable): String

os: OS

osVersion (nullable): String

osSupport (nullable): OsSupport

announcementUrl: String

lastMaintenanceDate: Date

lastVulnerabilityDate (nullable): Date

lastSupportDate: Date

components: Bag<DevicePart>

name: String

partId (nullable): String

serialNumber (nullable): String

partType: DevicePartType

description (nullable): String

versionId (nullable): String

support (nullable): DevicePartSupport

announcementUri: String

lastMaintenanceDate (nullable): Date

lastVulnerabilityDate (nullable): Date

lastSupportDate: Date

managementIps: Bag<IpAddress>

vIans: Bag<NamedVlanCollection>

interfaces: Bag<Interface>

networkInstances: Bag<NetworkInstance>

bgpRib (nullable): BgpRib

str: Str

Supported Platforms (Software and Hardware)🔗🔗

The complete list of platforms for which EoL/EoS lifecycle data is available is maintained internally and may grow over time.

OS lifecycle support coverage🔗🔗

Vendor	OS
Arista	EOS
Checkpoint	
Cisco	IOS
Cisco	IOS-XE
Cisco	IOS-XR
Cisco	NXOS
F5	

Vendor	OS
Fortinet	
Juniper	JUNOS
Palo Alto	CLOUDGENIX
Palo Alto Networks	PAN-OS

Hardware lifecycle support coverage

Vendor
CHECKPOINT
CISCO
F5
FORTINET
PALO_ALTO_NETWORKS

Summary

The NQE EoL/EoS data model provides a normalized, vendor-independent foundation for assessing lifecycle status across both software and hardware. It enables automated lifecycle evaluation in NQE queries and drives the Inventory+ OS/Hardware Support views, helping operators proactively manage lifecycle risk without external tooling.